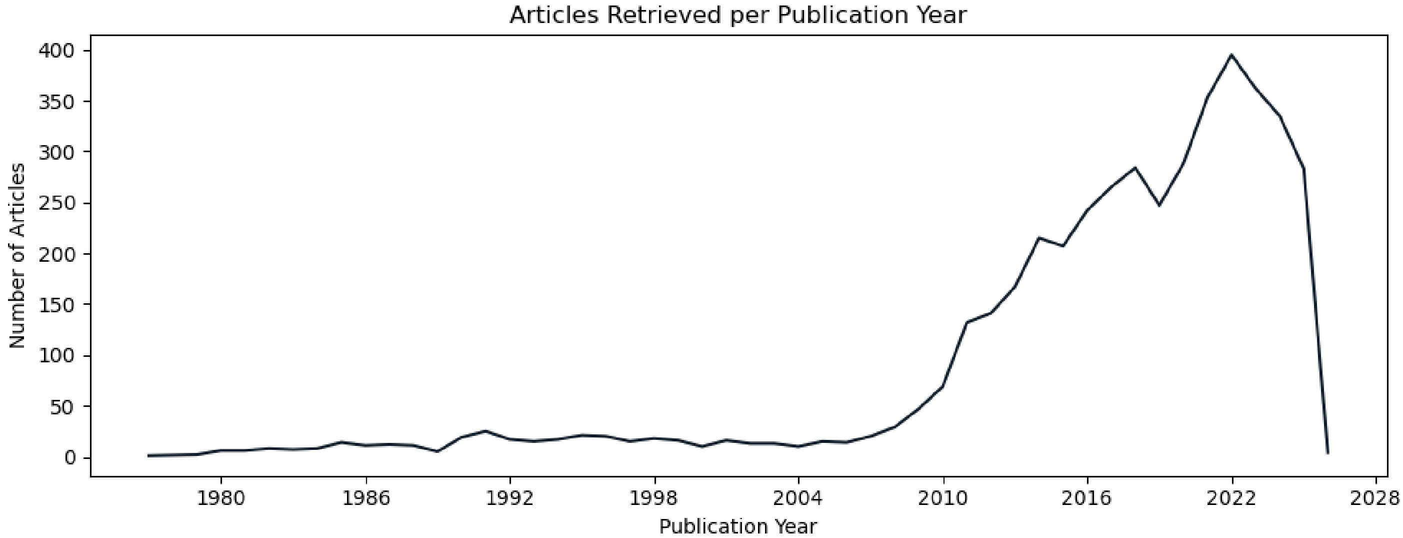
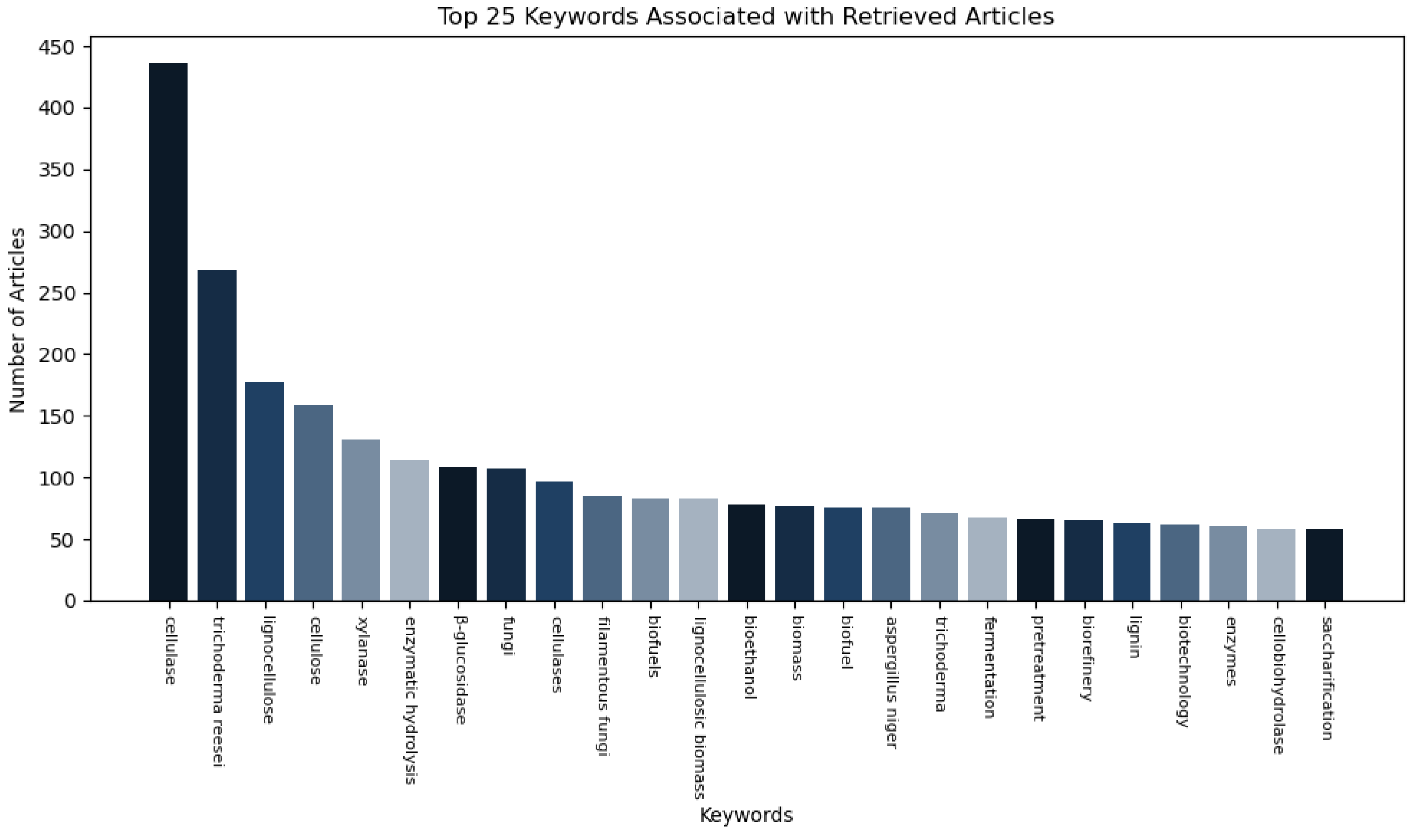
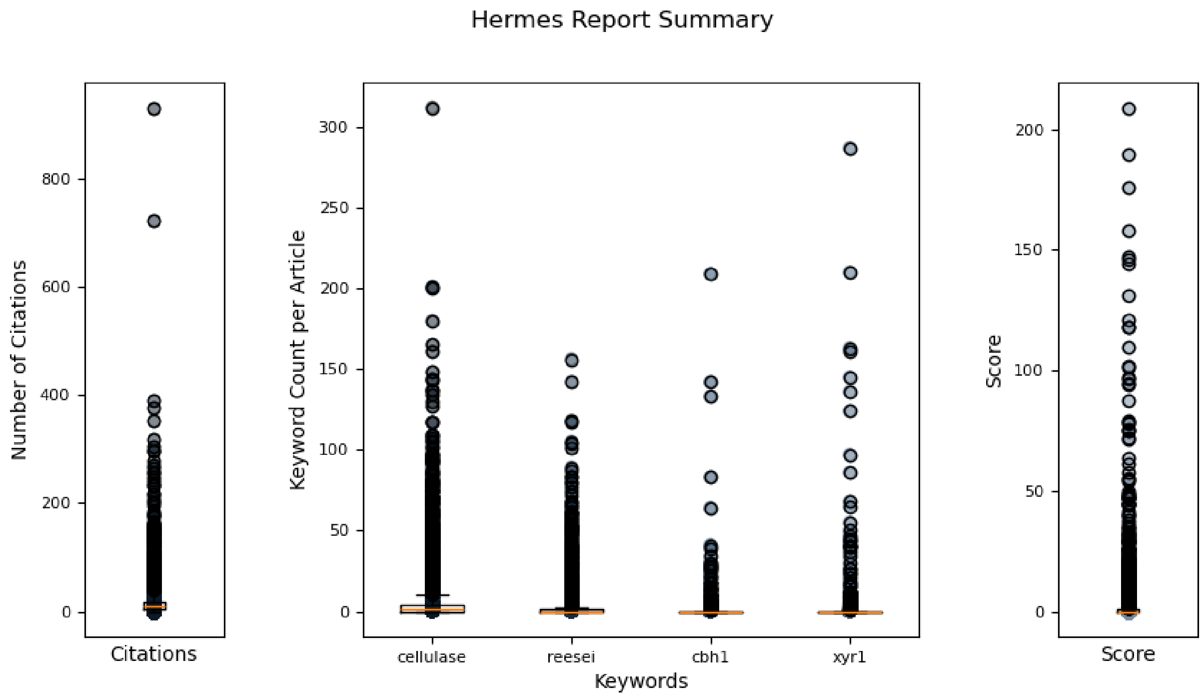
Literature Mining Report

Query Statistics

Query Date: 2025-12-19
Query Terms: "cellulase production trichoderma reesei"
Scoring Keywords: cbh1; cellulase; reesei; xyr1 (Default Mode)
Articles Parsed: 4461/4461 (100.00%)
Keyword Rarity Score (IDF): cellulase: 0.665; cbh1: 3.000; reesei: 1.015; xyr1: 3.285







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
26909077	2016	Frontiers in Mi	dos Santos Cast et al.	understanding the role of the master regulato...	52	209.09
25909478	2015	BMC Genomics	Lichius, Alexan et al.	genome sequencing of the trichoderma reesei q...	23	189.84
25302561	2014	Molecular Micro	Lichius, Alexan et al.	nucleo-cytoplasmic shuttling dynamics of the ...	40	175.90
22554051	2012	Molecular Micro	Seiboth, Bernha et al.	the putative protein methyltransferase lae1 c...	100	157.83
27462408	2015	Cell Discovery	Liu, Rui et al.	efficient genome editing in filamentous fungu...	160	147.03
24294104	2013	Current Genomic	Amore, Antonell et al.	regulation of cellulase and hemicellulase gen...	86	146.22
35317826	2022	Microbial Cell	Mattam, Anu Jos et al.	factors regulating cellulolytic gene expressi...	22	144.01
29229981	2017	Scientific Repo	Zheng, Fanglin et al.	the mating type locus protein mat1-2-1 of tri...	17	130.97
32765463	2020	Frontiers in Mi	Jiang, Xianzhan et al.	improved production of majority cellulases in...	7	121.01
29079620	2017	Applied and Env	Kiesenhofer, Da et al.	influence of cis element arrangement on promo...	19	118.06
32877410	2020	PLoS Genetics	Zheng, Fanglin et al.	trichoderma reesei xyr1 activates cellulase g...	18	118.05
36468854	2022	mSystems	Ma, Liang et al.	botrytis cinerea transcription factor bcxyr1 ...	8	109.88
31517564	2020	RNA Biology	Till, Petra et al.	regulation of gene expression by the action o...	6	101.72
22080343	2011	Applied Microbi	Uzbas, Fatma et al.	a homologous production system for trichoderm...	43	101.39
36376415	2022	Scientific Repo	Arai, Toshiharu et al.	inducer-free cellulase production system base...	6	97.00
24941042	2014	PLoS ONE	Silva-Rocha, Ra et al.	deciphering the cis-regulatory elements for x...	21	96.49
25386652	2014	PLoS ONE	Karimi Aghcheh, et al.	the velvet a orthologue vel1 of trichoderma r...	56	94.81
36528622	2022	Biotechnology f	Shen, Linjing et al.	tailoring the expression of xyr1 leads to eff...	3	94.37
38650205	2021	Bioresources an	Yan, Su et al.	from induction to secretion: a complicated ro...	20	87.82
34983541	2022	Microbial Cell	Wang, Yifan et al.	development of a powerful synthetic hybrid pr...	5	79.08
22750088	2013	Journal of Biot	Kubicek, Christ	systems biological approaches towards underst...	50	78.54
33606681	2021	PLoS Genetics	Wang, Lei et al.	interdependent recruitment of cyc8/tup1 and t...	11	78.24
23991059	2013	PLoS ONE	Wang, Mingyu et al.	a mitogen-activated protein kinase tmk3 parti...	44	75.77
28557371	2017	Microbial Biote	Druzhinina, Iri et al.	genetic engineering of trichoderma reesei cel...	79	75.04
26354856	2015	Biotechnology L	Jiang, Yanping et al.	enhancing saccharification of wheat straw by ...	8	74.84

Query Result 1/25

Understanding the Role of the Master Regulator XYR1 in Trichoderma reesei by Global Transcriptional Analysis

dos Santos Castro, Lilian; de Paula, Renato G.; Antonito, Amanda C. C.; Persinoti, Gabriela F.; Silva-Rocha, Rafael; Silva, Roberto N.

Frontiers in Microbiology

Year: 2016 PMID: 26909077 Citations: 52 Score: 209.09 DOI: 10.3389/fmicb.2016.00175

Keywords Hits: cellulase: 37, reesei: 65, cbh1: 0, xyr1: 145

Keywords Frequency (TF): cellulase: 0.004; reesei: 0.007; cbh1: 0.000; xyr1: 0.016

Abstract:

We defined the role of the transcriptional factorXYR1in the filamentous fungus Trichoderma reesei during cellulosic material degradation. In this regard, we performed a global transcriptome analysis using RNA-Seq of the xyr1 mutant strain of T. reesei compared with the parental strain QM9414 grown in the presence of cellulose, sophorose, and glucose as sole carbon sources. We found that 5885 genes were expressed differentially under the three tested carbon sources. Of these, 322 genes were upregulated in the presence of cellulose, while 367 and 188 were upregulated in sophorose and glucose, respectively. With respect to genes under the direct regulation of XYR1, 30 and 33 are exclusive to cellulose and sophorose, respectively. The most modulated genes in the xyr1 belong to Carbohydrate-Active Enzymes (CAZymes), transcription factors, and transporters families. Moreover, we highlight the downregulation of transporters belonging to the MFS and ABC transporter families. Of these, MFS members were mostly downregulated in the presence of cellulose. In sophorose and glucose, the expression of these transporters was mainly upregulated. Our results revealed that MFS and ABC transporters could be new players in cellulose degradation and their role was shown to be carbon source-dependent. Our findings contribute to a better understanding of the regulatory mechanisms of XYR1 to control cellulase gene expression in T. reesei in the presence of cellulosic material, thereby potentially enhancing its application in several biotechnology fields.

Summary:

Introduction: Our dependence on fossil fuels and concerns surrounding its impacts on the environment has generated worldwide interest in establishing new fuel and energy sources (Borin et al). Reduction in fossil fuel consumption using alternate energy sources is a major challenge that humankind will face in the coming decades. Bioethanol production using lignocellulosic biomass, however, is . Lignocellulo sic biofuels are among the most abundant renewable resources on the planet and is an alternative sustainable energy source for the production of second-generation biofuel.

Results: RNA sequencing data of T. reeseiQM9414 and the mutant strain xyr1were cultivated in cellulose, sophorose, and glucose as unique carbon sources and three biological replicates of each condition were submitted for RNA sequence analysis. Approximately 147 million of 100 bp paired-end reads were obtained for QM94 14 and 187 million for xxyr1, with a concordant pair alignment rate of 89.8 and 89,2, respectively. Regarding the number of nucleotides, the data correspond to 38 GB for Qm9414and 45 GBfor ixyr1. To confirm the deletion of the genexyre1in the mutant Xyr11, the files generated after the alignment of reads are visualized using IGV and the number o r reads mapping to gene expression level ofxyreas amounted to 122208.

Discussion: We aimed to achieve a comprehensive understanding of how gene expression is regulated by XYR1 in T. reesei in response to different carbon sources by RNA-seq and bioinformatics analysis. By comparing the growth in cellulose and sophorose, we identified new features in cellular degradation in T . eesi and showed a remarkable similarity between their transcriptomic profiles. The expression ofxyr1varies according to the analyzed condition. In cellulose, the role of the Xyr1, which is already well-established (Stricker et al.,) justifying its high level of gene expression in the parental strain QM9414. Conversely, we observed that in the presence of glucose, the expression ofxxyr was minimal (Figure), which may be related to the intervention of another transcription factor also involved in the regulation of cellulase expression in T, rese.

Conclusions: Our study contributes to a better understanding of the role of the transcription factor XYR1 in the degradation of cellulosic material by T. reesei. Our transcriptomic analysis showed that several genes have their expression affected in a carbon source-dependent manner. The transcriptional profile of xyr1was drastically altered during growth in the presence cellulose, sophorose, and glucose. The most differentially expressed genes include mainly cellulase enzymes, but expression of transporters and transcriptional factors are also affected by XyR1. Here, we highlighted the modulation of MFS and ABC family transporters. Since transport across the membrane is the first step at which nutrient supply is tightly regulated in response to intracellular needs and often also the rapidly changing external environment, the study of these transporters will contribute to the understanding of molecular mechanisms underlying the regulation of cellulose enzyme synthesis in this fungus. Furthermore, these data will contribute the construction of industrial strains of T.

Figures:

Figure 1: RNA-seq coverage plots of the xyr1 gene . Number of sequence reads (x-axis) along the genomic region corresponding to the xyr1 gene (Protein ID 122208, scaffold 11, genomic position 201774204723).

Figure 2: Gene Regulatory Network (GRN) of 1184 differentially expressed genes in xyr1 /QM9414 in response to different carbon sources (cellulose,

sophorose, and glucose) . Genes are represented as nodes (shown as circles), and interactions are represented as edges (red lines: up-regulated interactions; and green lines: down-regulated interactions), connecting the nodes: 1674 interactions.

Figure 3: GO enrichment analysis of up- and down-regulated genes of the xyr1 strain compared to parental strain QM9414 grown in cellulose, sophorose, and glucose as sole carbon sources . The enriched GO terms according to molecular function, cellular component, and biological process in T. reesei . Significantly enriched categories (P 0.05) are shown.

Figure 4: Gene expression profiles of Carbohydrate Active enZymes (CAZy) of the xyr1 strain compared to parental strain QM9414 . Fragments per kilobase of exon per million fragments mapped (FPKM) for each enzyme class cultured in presence of glucose, cellulose, and sophorose. (A) All CAZymes identified in this study; (B) Only the differentially expressed CAZymes of each carbon source were used in this analysis. AA, Auxiliary Activitiesredox enzymes that act in conjunction with CAZymes; CBM, Carbohydrate-Binding Modulesadhesion to carbohydrates; CE, Carbohydrate Esteraseshydrolysis of carbohydrate esters; GH, Glycoside Hydrolaseshydrolysis and/or rearrangement of glycosidic bonds; GT, GlycosylTransferasesformation of glycosidic bonds; PL, Polysaccharide Lyasesnon-hydrolytic cleavage of glycosidic bonds.

Figure 5: Maximum-likelihood phylogenetic tree of transporters genes identified exclusively in the conditions studied . The radial tree was generated using iTOL (Letunic and Bork, 2007 , 2011).

Figure 6: Scheme involving the direct regulation by the Zn2Cys6 transcription regulator XYR1 . This figure shows the processes that are expected to be involved in xyr1 regulation of lignocelluosic enzymes in T. reesei . Endoglucanases (EGs) hydrolyze cellulose bonds internally, while cellobiohydrolases cleave cellobiose units from the ends of the polysaccharide chains. The released cellobiose units (disaccharides) are subsequently hydrolyzed by -glucosidases, releasing glucose, the main carbon source readily metabolisable by T. reesei . Expansin proteins (Swollenin) rapidly induce extension of plant cell walls by weakening the noncovalent interactions that help to maintain their integrity. Cellobiose dehydrogenase (CDH) is a potential electron donor for polysaccharide monooxygenases (PMOs). EGs and PMOs internally cleave cellulose chains releasing chain ends that are targeted by cellobiohydrolases (CBHs). Dotted arrows indicate potential areas of research for enhancement of enzymes secretion in T. reesei : XYR1 activates or represses (directly) the transcription of genes in cellulose and sophorose. PWM XYR1 defined in Silva-Rocha et al. (2014).

Mentioned biomedical entities:

Genes: 3, ABC, ACE1, ACE2, ACE3, Ace2, AraR, Aro, BXL1, CA, CBM1, CLP1, CRE1, Ctr1, FPA, GH, GH31, GT, Hap2, Hxt, LD, Ld, MFS, Myb, PL, XYR1, XlnR, actin, cellulase, myb, swo1, xyn2, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: A. nidulans, Escherichia coli, human

Tissues: tube

Pathways: N/A

Query Result 2/25

Genome sequencing of the Trichoderma reesei QM9136 mutant identifies a truncation of the transcriptional regulator XYR1 as the cause for its cellulase-negative phenotype

Lichius, Alexander; Bidard, Frdrique; Buchholz, Franziska; Le Crom, Stphane; Martin, Joel; Schackwitz, Wendy; Austerlitz, Tina; Grigoriev, Igor V; Baker, Scott E; Margeot, Antoine; Seiboth, Bernhard; Kubicek, Christian P

BMC Genomics

Year: 2015 PMID: 25909478 Citations: 23 Score: 189.84 DOI: 10.1186/s12864-015-1526-0

Keywords Hits: cellulase: 49, reesei: 42, cbh1: 2, xyr1: 210

Keywords Frequency (TF): cellulase: 0.007; reesei: 0.006; cbh1: 0.000; xyr1: 0.031

Abstract:

Trichoderma reesei is the main industrial source of cellulases and hemicellulases required for the hydrolysis of biomass to simple sugars, which can then be used in the production of biofuels and biorefineries. The highly productive strains in use today were generated by classical mutagenesis. As byproducts of this procedure, mutants were generated that turned out to be unable to produce cellulases. In order to identify the mutations responsible for this inability, we sequenced the genome of one of these strains, QM9136, and compared it to that of its progenitor T. reesei QM6a.

Summary:

Background: Public concerns about the consequences of the continued production of fuels and chemicals from fossil carbon sources have today led to intensive attempts to switch to the use of renewable carbon resources for future energy supply, such as lignocellulosic biomass from agricultural crop residues, grasses, wood and municipal solid waste. This affects not only the production of ethanol as a biofuel but also a range of key biochemical building blocks for biorefineries, including 1,4-dicarboxy acids, 3-hydroxypropionic acid, levulinic acid, and polyols; to only name a few. Thereby, the costs of cellulases and hemicellulasses needed for hydrolysis of the plant biomass to soluble substrates contribute substantially to the price, and much cheaper sources of these enzymes are urgently required.

Results: The strain QM9136 has been isolated by a single round of X-ray mutagenesis using a linear particle accelerator from the wild type strain Q6a and was identified to be unable to produce cellulases . To get the complete picture of genetic modifications in that strain, we conducted whole genome comparisons between the natural isolate, the QM6a strain, and the mutant strain Qm9136 which we sequenced using high-throughput Illumina sequencing. To ...

Conclusion: The missing 140 C-terminal amino acids of XYR1 are therefore responsible for its previously observed auto-regulation which is essential for cellulases to be expressed. Our data present a working example of the use of genome sequencing leading to a functional explanation of the QM9136 cellulosae-negative phenotype. Electronic supplementary materialThe online version of this article (doi:10.1186/s12864-015-1526-0) contains supplementary materials, which are available to authorized users.

Discussion: Historically, the cloning of genes was based on the use of complementation of mutants, which favored those organisms which either contained plasmids or autonomously replicating elements (for easy recovering of the target gene) or which could efficiently be crossed in the laboratory. Although a number of substitute strategies had been developed (cf.), this nevertheless strongly limited the progress in organisms that exhibited neither of these possibilities, such as the so-called imperfect fungi. The advances in sequencing technologies and bioinformatics have now made it possible to sequence even full eukaryotic genomes quickly, and to use this strategy to identify mutations responsible for a particular phenotype -. However, since mutant organisms often contain several mutations (cc.) , most researchers either mapped the target mutation or backcrossed their mutant against the wild type, and sequenced more than one mutant genome simultaneously. Since the T. reeseimutants were maintained asexually, neither of those two strategies was possible for the present study.

Conclusions: Our study demonstrates how high-throughput genomic sequencing combined with reverse genetics can provide novel functional insights into organisms that are otherwise only poorly susceptible to mutational analyses. The mutation in xyr1inT. reeseiQM9136 also allowed new insights into the function of this central cellulase regulator. The ongoing investigation of yet two more cellullase-negative strains of T. reaesi (i.e. QM9978 and QM 9979;) which bear a functional xir1copy is expected to further elucidate celluluse and hemicellulases regulation in this important filamentous fungus.

Figures:

Figure 1: Pedigree of the cellulase-negative strain QM9136. UV light and chemical mutagenesis programs of the NATICK 3 , 4 , 9 and RUTGERS 27 series lead to the selection of T. reesei strains with improved (+) or impaired () cellulase production. Adapted with permission from 2 .

Figure 2: XYR1 structure and structural consequences of the truncated XYR 1780 . (A) Predicted coiled-coil domains in XYR1 using COILS, which compares a sequence to a database of known parallel two-stranded coiled-coils and derives a similarity score from which the probability that the sequence will adopt a coiled-coil confirmation is calculated. (B) XYR1 contains a classical tri-partite nuclear localisation signal (NLS; green) which

directs the transcription factor into the nucleus. XYR1s DNA-binding domain (DBD; yellow) is orthologous to that of Gal4 from Saccharomyces cerevisiae . The first coiled-coil domain of XYR1 (orange) is located at the end of the DBD. The filamentous fungal-specific transcription factor domain (FSTFD; blue) regulates target gene expression and constitutes the biggest part of the protein. A second coiled-coil domain (orange) overlaps with the C-terminal end of the FSTFD. The acidic activation domain (AAD; dark red) is a characteristic feature of Zn 2 Cys 6 -type transcription factors. The third coiled-coil domain (orange) sits at the very C-terminus, and likely mediates homodimerisation of XYR1. Hence, the truncated XYR1 1780 protein encoded in QM9136 almost completely lacks the AAD and fully lacks the third coiled-coil domain due to a frame-shift mutation from L765 onwards. The terminal 16 amino acids (L765-F780; light red) encode a non-sense sequence that finds no BLASTp hit in the T. reesei genome. (C) 3D-reconstruction of XYR1. RaptorX protein structure prediction software (raptorx.uchicago.edu) identifies three functional domains in XYR1, resembling DNA-binding domain (DBD), fungal-specific transcription factor domain (FSTFD) and acidic-activation domain (AAD), perfectly matching our manual annotation (B) . Five of the seven -helices comprising the AAD are missing in the truncated XYR1 1780 protein (highlighted in blue). The closest structural orthologs are the Zn 2 Cys 6 -type transcriptional regulators Ppr1 and Gal4 from S. cerevisiae , both known to be active as non-symmetrical homodimers 63 .

Figure 3: Gene replacement with full-length GFP-XYR1 construct rescues XYR1-loss-of-function phenotype of QM9136. (A) QM9136::GFP-XYR1 transformants (TRAL008 clones 1 to 3) grow on 65mM D-xylose show colony extension speeds comparable to the moderate producer strain QM9414 and QM9414::GFP-XYR1 transformants (TRAL002 clones 1 and 2). QM9136 and xyr1 are sensitive to 65mM D-xylose and display significantly reduced colony extension speeds. (B) Macroscopic colony phenotypes of GFP-XYR1 transformants TRAL002 and TRAL008, and control strains after 5days on MA medium supplemented with 65mM D-xylose. The XYR1-loss-of-function strains xyr1 and QM9136 develop a slower and less dense growing mycelium, however, start conidiation earlier, compared to the reference strain QM9414. Expression of GFP-XYR1 restores wild type morphology in QM9136 but does not alter QM9414. (C) Gene expression of the major cellulase cel7a , xylanase xyn2 and transcription factor xyr1 itself, induced by one hour exposure to 1.4mM sophorose, are restored in QM9136::GFP-XYR1 transformants (TRAL008). The XYR1-loss-of-function strains QM9136 and xyr1 show now transcriptional response to induction by sophorose. Both cellulase-positive controls, QM9414 and QM9414::GFP-XYR1 transformants (TRAL002) respond as expected with upregulated cel7a , xyn2 and xyr1 expression. Error bars show standard deviations (n=6 from three experimental and two biological replicates).

Figure 4: Gene replacement with the truncated GFP-XYR1 1780 reproduced the XYR1-loss-of-function phenotype in QM9414. (A) QM9414::GFP-XYR1 1780 transformants (TRAL007 clones 1 to 3) showed the same growth defect on 65mM D-xylose as QM9136 and xyr1 , respectively, indicating loss of XYR1 function, which was further verified by the corresponding macroscopic colony phenotypes, shown in (B) . (C) Gene expression of cellulase cel7a , xylanase xyn2 and xyr1 were not significantly upregulated in GFP-XYR1 1780 expressing transformants (TRAL007), whereas the GFP-XYR1 transformants (TRAL002) clearly responded to the inductive stimulus provided by sophorose. Error bars show standard deviation (n=6 from three experimental and two biological replicates involving two different TRAL007 clones).

Figure 5: Nuclear import of GFP-XYR1 is indistinguishable in QM9136 and QM9414 transformants. (A) Sophorose-induced nuclear import of GFP-XYR1 in QM9136 and QM9414 backgrounds was indistinguishable. Relative fluorescence intensity represents the average GFP-XYR1 signal detected measured in 120 nuclei per sample. The nucleo-cytoplasmic ratio (n/c-ratio) provides a measure of how much brighter the average GFP-XYR1 signal is inside nuclei compared to the surrounding cytoplasm. Error bars represent standard deviation (n=120). (B) Representative, inverted fluorescence images showing increasing GFP-XYR1 recruitment into nuclei of T. reesei germlings over a time course of 60min after carbon source replacement with 1.4mM sophorose.

Figure 6: Western blot analysis of GFP-XYR1 and GFP-XYR1 1780 under cellulase inducing conditions. Top row: Coomassie stained loading control; middle row: chemifluorescent image of Western blot membrane; bottom row: x-ray image of Western blot membrane. (A) With time (1-3h of sophorose induction), less of the 127kDa full-length GFP-XYR1(arrowheads) but more of its various 30-70kDa degradation products (asterisks) can be detected in protein extracts. This rapid turn-over of the transcription factor is conserved in QM9414 and QM9136 transformants. (B) In comparison to full-length GFP-XYR1 (arrowheads) expressed in QM9414 and QM9136 transformants, of the truncated GFP-XYR1 1780 construct, only 80kDa or smaller degradation products could be detected (asterisk). 5-30g refer to the total protein load per lane; numbers on the side denote molecular weight ranges; M1 is PageRuler pre-stained molecular weight marker; M2 is SuperSignal enhanced protein ladder; 18h QM6a GFP is the positive control, and 18h QM9414 the negative control for GFP(B-2) antibody specificity.

Figure 7: C-terminal truncation renders GFP-XYR1 1780 non-responsive to inductive signals. (A) Nuclear recruitment dynamics assessed by quantitative live-cell imaging. In contrast to the full-length GFP-XYR1 (TRAL002), whose concentration in nuclei rapidly increased by a factor of up to eight-fold, the truncated GFP-XYR1 1780 (TRAL007) was unable to significantly increase its presence in the nucleus in response to cellulase- and xylanase-inducing carbon sources (1.4mM sophorose, 25mM lactose and 1mM D-xylose). Instead, GFP-XYR1 1780 consistently showed nuclear presence comparable to the basal presence of GFP-XYR1 under xyr1 -non-inducing conditions (65mM D-xylose). Error bars show standard deviation (n=2 from two biological replicates involving two different TRAL007 clones). (B) Representative, inverted fluorescence images showing nuclear import of GFP-XYR1 and GFP-XYR1 1780 after one hour of sophorose induction. In contrast to the full-length GFP-XYR1 construct, the truncated version is

barely visible inside nuclei before induction (arrows). Most importantly, upon addition of 1.4mM sophorose, the nuclear signal of GFP-XYR1 1780 does increase, however, only to a fraction of the intensity (about 15%) of the GFP-XYR1 construct under identical conditions. Scale bar, 10m.

Figure 8: Quantification of nuclear recruitment of GFP-XYR1 and GFP-XYR1 1780 in the three main functional zones of the mature colony. All strains were grown for 48hours on MA medium agar plates supplemented with the indicated carbon source. (A) Full-length GFP-XYR1 responds to different xyr1 -inducing signals with nuclear recruitment, most prominently in the central area of the colony, and dependent on the inductive strength of the provided carbon source (25mM lactose1mM D-xylose1% (w/v) cellulose). Under non-inducing/repressing conditions (50mM D-glucose or 65mM D-xylose) only weak baseline nuclear recruitment can be detected. In contrast, the truncated GFP-XYR1 1780 does not significantly respond to cellulase induction, and hence can only be detected at baseline levels independent of the provided carbon source. (B) Representative, inverted fluorescence images of the tested conditions, showing that GFP-XYR1 1780 does respond to 25mM lactose with weak but detectable nuclear import, comparable to the baseline presence of GFP-XYR1 under repressing conditions (barely visible nuclei are indicated with arrows). On 50mM D-glucose GFP-XYR1 1780 is visible in small spots, suggesting degradation of the non-functional XYR1 construct in the proteasome.

Mentioned biomedical entities:
Genes: ACE2, ACE3, AM, B, CBM1, CDC1, CRE1, ENS, Gal4, MA, Ppr1, RUT, SEB, SRA, XYR1, XlnR, Xyr1, cbh1, cel6a, cellulase, cre1, ptrA, tef1, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: GE

Tissues: lens, tube

Pathways: N/A

Query Result 3/25

Nucleo-cytoplasmic shuttling dynamics of the transcriptional regulators XYR1 and CRE1 under conditions of cellulase and xylanase gene expression in Trichoderma reesei

Lichius, Alexander; Seidl-Seiboth, Verena; Seiboth, Bernhard; Kubicek, Christian P

Molecular Microbiology

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Keywords Hits: cellulase: 74, reesei: 29, cbh1: 9, xyr1: 163

Keywords Frequency (TF): cellulase: 0.009; reesei: 0.003; cbh1: 0.001; xyr1: 0.019

Abstract:
Trichoderma reesei is a model for investigating the regulation of (hemi-)cellulase gene expression. Cellulases are formed adaptively, and the transcriptional activator XYR1 and the carbon catabolite repressor CRE1 are main regulators of their expression. We quantified the nucleo-cytoplasmic shuttling dynamics of GFP-fusion proteins of both transcription factors under cellulase and xylanase inducing conditions, and correlated their nuclear presence/absence with transcriptional changes. We also compared their subcellular localization in conidial germlings and mature hyphae. We show that cellulase gene expression requires de novo biosynthesis of XYR1 and its simultaneous nuclear import, whereas carbon catabolite repression is regulated through preformed CRE1 imported from the cytoplasmic pool. Termination of induction immediately stopped cellulase gene transcription and was accompanied by rapid nuclear degradation of XYR1. In contrast, nuclear CRE1 rapidly decreased upon glucose depletion, and became recycled into the cytoplasm. In mature hyphae, nuclei containing activated XYR1 were concentrated in the colony center, indicating that this is the main region of XYR1 synthesis and cellulase transcription. CRE1 was found to be evenly distributed throughout the entire mycelium. Taken together, our data revealed novel aspects of the dynamic shuttling and spatial bias of the major regulator of (hemi-)cellulase gene expression, XYR1, in T. reesei.

Summary:
Introduction: In nature fungi contribute as essential decomposers of complex organic molecules to the biological carbon cycle. These organic polymers, produced through carbon dioxide fixation by plants, mainly comprise plant cell wall polysaccharides including cellulose, hemicelluloses, pectins and the polymer lignin. The major plant cellwall component is the -(1,4)-linked glucose polymer, cellulose that alone exhibits an annual production of 7.2 10 10 tons, and its microbial degradation is therefore a key transformation step in the biological Carbon Cycle.

Results: XYR1 requires N-terminal and CRE1 requires C-terminum GFP labeling in order to preserve protein function. Consequently, only T. reeseitransformants expressing GFPXYr1 and Cre1GFP, respectively, were used for all subsequent experiments. Xyr1 requires GFP and CRe1 requires c-terminus GFP. xyr1/cre1

Discussion: It is commonly believed that most Zn2Cys6zinc cluster proteins are localized within the nucleus on a constitutive and permanent basis (MacPherson However, there is now an increasing number of these zinc cluster proteins from multicellular fungi known, which do not share this behavior (Shimizu Berger Makita Dinamarco Here we showed that the XYR1 transcription factor ofT. reeseialso joins this list.

Figures:
Figure 1: Nuclear import of XYR1 was significantly increased under cellulase inducing conditions.A and B. (A) GFPXYR1 became strongly recruited into nuclei in the presence of lactose, whereas it was only weakly expressed under repressing/non-inducing conditions on glucose (B).C and D. CRE1 did not show this specific response and appeared equally recruited to nuclei under cellulase inducing and repressing conditions. 3D surface plots (right row) of the inverted fluorescence images (middle row) illustrate the relative differences of localized fluorescence signal intensities. The corresponding transmission light images are shown in the left row. The displayed strains were incubated for 48h at 28C on MA agar plates supplemented with lactose or glucose respectively.Scale bars, 5m.

Figure 2: Protein turn-over kinetics of both fusion reporters differ, leading to significant nuclear accumulation of free GFP from GFPXYR1 degradation.A. Coomassie-stained SDS-PAGE gel. Except for the GFP positive control, of which only 0.35g total protein extract of QM6a P tef1 :: gfp were loaded, 20g of all other protein extract samples were loaded per lane. Molecular weight marker on both sides is PageRuler 10170kDa ladder.B. Western blot of the sister SDS-PAGE gel shown in (A), incubated with monoclonal -GFP(B-2) antibody, ECL 2 reagent and imaged on a Typhoon FLA700 chemifluorescence imager. Molecular weight marker on the left (lane 1) is PageRuler 10170kDa ladder, on the right (lane 8) is SuperSignal Enhanced 20 150kDa ladder. Cytosolic GFP (26.98 kDa) as positive control is exclusively detected in lane 2. Full-length CRE1GFP (71.07kDa) bands in lane 3 just above the 70kDa marker, and some of its degradation products with decreasing abundance towards low-molecular-weight species below. Lanes 46 show full-length GFPXYR1 (128.41kDa) with decreasing abundance after 13h of sophorose induction. Interestingly, with time, abundance of its higher-molecular-weight degradation products (4080kDa) decreases, while amounts of its low-molecular-weight degradation products (25 30kDa) significantly increase. The negative control in lane 7 (protein extract from an 18h glucose culture of QM9414 t ku70) reveals minor non-specific background detection in the higher molecular weight range40kDa.C. The same membrane as shown in (B) developed by x-ray film exposure, shows identical results with less non-specific background signals, however, inferior spatial resolution. Intense nuclear fluorescence of GFPXYR1 and

CRE1GFP, respectively, was confirmed by live-cell imaging in all samples prior to biomass harvest and protein extraction.

Figure 3: XYR1 and CRE1 showed distinct nuclear shuttling dynamics. Inverted fluorescence images show the degree of nuclear transcription factor recruitment at representative time points, with the corresponding quantitative analysis below.A. Time-course of sophorose-induced nuclear import of GFPXYR1. Quantitative image analysis showing immediate but weak nuclear recruitment which started to significantly increases after 30min upon induction, probably correlated with the maturation time of newly synthetized GFP.B. Time-course of glucose-induced nuclear loss of GFPXYR1. Quantitative image analysis showing only slow decrease of nuclear XYR1 signal within the first hour upon carbon source replacement.C. Time-course of glucose-induced nuclear import of CRE1GFP. Quantitative image analysis showing rapid nuclear import of CRE1 from the cytoplasmic pool.D. Time-course of sophorose-induced nuclear loss of CRE1GFP. Quantitative image analysis showing rapid nuclear export of CRE1 upon cellulase induction with sophorose. Please refer to the text for more detailed explanation.Scale bars, 5m.

Figure 4: XYR1 nuclear import dynamics correlated with gene expression.A. Sophorose-induced nuclear import of XYR1 represents cellulase inducing conditions.B. Low-xylose (1mM) induced nuclear import of XYR1 represents cellulase and xylanase inducing conditions.C. High-xylose (65mM) induced XYR1 suppression represents carbon-catabolite repressing conditions.D. Starvation-induced nuclear import of XYR1 represents carbon-catabolite de-repressing conditions.E. Gene expression analysis from biomass samples collected at the end of each imaging experiment quantified in (AD) showed that upon sophorose addition cellulase cbh1 and xyr1 itself were strongly upregulated, low concentration (1mM) of xylose strongly induce cbh1 and xyn2 , but not xyr1 , whereas in the presence of 65mM xylose none of these three marker genes was significantly upregulated. This is different under de-repressing conditions, i.e. in the absence of any experimentally added carbon source, where nuclear import of XYR1 evidently induced moderate upregulation of cbh1 , xyn2 and of itself.

Figure 5: Complete inhibition of XYR1 nuclear import and cellulase gene expression by cycloheximide treatment. (A)(D) show the shuttling dynamics of fluorescently labeled XYR1 and CRE1 upon cellulase induction and repression, in the presence and absence of 50M cycloheximide (CHX) respectively. (E) and (F) show the corresponding transcriptional analyses.A. Addition of CHX completely blocked de novo protein biosynthesis of XYR1 and consequently its nuclear import.B. Nuclear loss of CRE1 remained unaffected under the same conditions.C. Nuclear loss of XYR1, on the other hand, was slightly enhanced in the presence of CHX in comparison to the untreated control.D. Glucose-induced nuclear import of CRE1 showed no significant differences between CHX-treated sample and untreated control.E. Inhibition of de novo XYR1 protein biosynthesis and nuclear import was directly correlated to the inhibition of transcription of XYR1-dependent genes (cbh1 and xyn2). Transcription of xyr1 itself, however, responded normally to the induction signal and became upregulated to the same extend as in the untreated controls.F. Apart from a weak upregulation of cre1 in the presence of CHX, all other tested marker genes became downregulated as expected under cellulase non-inducing conditions.

Figure 6: Transcription factor recruitment in the functionally stratified colony.A. Forty-eight-hour lactose plate culture of a GFPXYR1 expressing strain. The white dotted rectangle indicates the sampling area which includes all three main functional zones of the colony (central area, subperiphery and leading edge) assessed by live-cell imaging microscopy.B. Bright-field images show the typical hyphal phenotypes: dense hyphal network with conidiation (conidia indicated with asterisks) and a high degree of vacuolation (vacuoles indicated with arrowheads) in the central area, branched hyphae with beginning vacuolation but no conidiation in the subperiphery, and largely non-vacuolated leading hyphae with clear apical nuclear-exclusion zone (indicated with black lines). Representative images of the GFPXYR1 fluorescence signal from these three areas show the evident decrease in nuclear XYR1 signal from colony center to colony edge. Scale bars, 5m.C and D. Quantification of nuclear recruitment of XYR1 and CRE1, respectively, in these three zones on inducing (cellulose and lactose) and non-inducing/repressing (glucose and xylose) carbon sources showed that nuclear import of XYR1 peaked in the central colony area exclusively under cellulase inducing conditions. CRE1, on the other hand, was equally recruited to nuclei in all regions of the colony with no significant differences between the tested carbon sources. All cultures have been incubated for 48h at 28C on MA medium supplemented with 1% w/v of the respective carbon source.

Mentioned biomedical entities:

Genes: ACE1, ACE2, AflR, CRE1, Cellulase, Cre1, CreA, Gal1, Gal4, Leu3, Lys14, MA, Mig1, Oaf1, Put3, Snf1, Tel, War1, XYR1, XlnR, Xyr1, cbh1, cellulase, cre1, gpd1, hydrolase, pkiA, tef1, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: A. nidulans, E. coli, GE

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 4/25

The putative protein methyltransferase LAE1 controls cellulase gene expression in Trichoderma reesei

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Molecular Microbiology

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Keywords Frequency (TF): cellulase: 0.013; reesei: 0.012; cbh1: 0.000; xyr1: 0.004

Abstract:

Trichoderma reesei is an industrial producer of enzymes that degrade lignocellulosic polysaccharides to soluble monomers, which can be fermented to biofuels. Here we show that the expression of genes for lignocellulose degradation are controlled by the orthologous T. reesei protein methyltransferase LAE1. In a lae1 deletion mutant we observed a complete loss of expression of all seven cellulases, auxiliary factors for cellulose degradation, -glucosidases and xylanases were no longer expressed. Conversely, enhanced expression of lae1 resulted in significantly increased cellulase gene transcription. Lae1-modulated cellulase gene expression was dependent on the function of the general cellulase regulator XYR1, but also xyr1 expression was LAE1-dependent. LAE1 was also essential for conidiation of T. reesei. Chromatin immunoprecipitation followed by high-throughput sequencing (ChIP-seq) showed that lae1 expression was not obviously correlated with H3K4 di- or trimethylation (indicative of active transcription) or H3K9 trimethylation (typical for heterochromatin regions) in CAZyme coding regions, suggesting that LAE1 does not affect CAZyme gene expression by directly modulating H3K4 or H3K9 methylation. Our data demonstrate that the putative protein methyltransferase LAE1 is essential for cellulase gene expression in T. reesei through mechanisms that remain to be identified.

Summary:

Introduction: Most industrial production of enzymes for plant biomass hydrolysis is performed with mutants of the fungus Trichoderma reesei(the anamorph of the tropical ascomyceteHypocrea jecorina). Consequently, this fungi serves as the model system for the molecular understanding of cellulase gene expression and secretion of the encoded cellullase proteins. To this end, its genome has recently been sequenced.

Results: Identification of the T. reeseiLAE1 orthologue. Phylogenetic analysis of LaeA protein sequences produced a tree. 2011 American Chemical Society. All rights reserved. - 2012 American Chemical society. All Rights Reserved. ---------- ---------- --- 2012 American Chemical Societies. All Right Reserved --- -------- -- ---2012 American Chemicalsocieties - All Rights reserved.

Discussion: The methyltransferase LAE1 influences cellulasa gene transcription in T. reesei, and may thus represent a novel approach for increasing total cellulosas activity and strain improvement by recombinant techniques in this fungus. Although the use of LaeA for increasing secondary metabolite production by fungi has been proposed we are not aware of any successful demonstration of this principle in an industrially relevant fungal species.

Figures:

Figure 1: Phylogenetic analysis of LaeA/LAE1 proteins from Eurotiomycetes , Dothidiomycetes and Sordariomycetes. Accession numbers for protein sequences are listed in Table S3. The tree was constructed by Neighbor Joining in MEGA 5.0 (Tamura et al ., 2011) with 500 bootstrap replicates (coefficients are indicated below the respective nodes). Gaps in the alignment were not considered.

Figure 2: Effect of loss-of-function of lae1 on biomass formation and cellulase/hemicellulase enzyme formation by T. reesei. Growth (A) and cellulase formation (B) of T. reesei QM 9414, the transformation recipient ku70 and the corresponding lae1 strains CPK3793 and CPK3791 on 1% (w/v) lactose. Cellulase expression is given in arbitrary units and related to the respective biomass dry weight of the strain at the respective time point (given in A). The three bars represent (from left to right) values for 48, 72 and 96 h of cultivation. Experiments are means of three biological replicates, and the SD given by vertical bars.

Figure 3: Growth of T. reesei QM 9414 (blue), the lae1 strain (CPK3793, red) and the strain expressing tef1:lae1 (CPK4086, green) on several oligosaccharides. Data were obtained from Phenotype microarrays as described (Druzhinina et al ., 2006). Carbon sources used were: (A) -cyclodextrin, (B) -cyclodextrin, (C) maltose, (D) d -melizitose - d -glucopyranosyl-(13)- O -- d -fructofuranosyl-(21)-- d -glucopyranoside, (E) -methyl-- d -galactoside, (F) lactulose 4- O - - d -galactopyranosyl- d -fructofuranose, (G) palatinose 6- O -- d -Glucopyranosyl- d -fructose, (H) stachyose - d -Fructofuranosyl- O -- d -galactopyranosyl-(16)- O -- d -galactopyranosyl-(16)-- d -glucopyranoside. The vertical axis shows the OD 750 that is equivalent to biomass formation (g l 1).

Figure 4: Expression of the two cellobiohydrolase-encoding genes cel7a and cel6a in T. reesei QM 9414 and the lae1 strain CPK3793 during growth on lactose or incubation with sophorose. Cellulase transcript levels in QM 9414 during growth on lactose are given with full bars and set to 1.0. The respective transcript levels in relation to QM 9414 are shown with open bars. Data are means of triplicate determinations from two biological replicates.

Figure 5: Biomass formation (A), cellulase production (B) and extracellular protein (C) during growth of T. reesei QM 9414 (QM) and several mutant strains bearing additional copies of the tef1:lae1 gene construct (CPK3791, CPK4326, CPK4086, CPK4087, CPK4325) on lactose. The three bars represent (from left to right) values for 48, 72 and 96 h of cultivation. Each bar is from a single experiment only but representative of at least four biological replica that were consistent with the claims.

Figure 6: Relative abundance of transcripts for cel7a, cel6a and lae1 at 26 h of growth on lactose in two T. reesei tef1:lae1 mutant strains (CPK4086, grey bars; CPK3791, white bars) in relation to the QM 9414 recipient strain (black bars). Transcripts were normalized to the housekeeping gene tef1 , and the respective ratio in QM 9414 set to 1. Other strains are given in percentage to that of the QM 9414. Copy numbers were determined as described in Experimental procedures : CPK4086 had two additional copies indicating a single ectopically integrated copy; CPK3791, however, had three additional copies, suggesting more than one ectopically integrated copy, but the exact number was not determined. Vertical bars indicate SD.

Figure 7: Cellulase activity during growth of selected T. reesei tef1:lae1 mutant strains on cellulose. Activities are given as arbitrary units and are related to 1 g of fungal biomass protein. Bars indicate measurements after 7, 9 and 11 days of incubation (from left to right). Data are means from four measurements and two independent biological replicates.

Figure 8: Relative changes in expression of CAZome genes in the lae1 (CPK3793) and the tef1:lae1 (CPK4086) strains, given as the fold changes of transcript hybridization in relation to the parent strain. Only values with P 0.05 are shown.

Figure 9: Cellulase formation in delta- lae1 (CPK3793) and delta- xyr1 strains of T. reesei is unaffected by the constitutive overexpression of xyr1 or lae1 . A. Overexpression of xyr1 in a delta- lae1 background: three different transformants yielded identical results and thus shown only for one of them (circles). The wild-type QM 9414, and QM 9414 containing a single pki1:xyr1 copy are given as comparison (diamonds and squares respectively). B. Overexpression of lae1 in a delta- xyr1 background: three different transformants yielded again identical results and thus shown only for one of them (circles). The wild-type QM 9414, and strain CPK4086 overexpressing tef1:lae1 are given as comparison (diamonds and squares respectively). Data from a single experiment only are shown, but are consistent with the results from at least two separate biological replicas.

Figure 10: Selected results of ChIP-seq with antibodies against H3K9me3, H3K4me3 and H3K4me2 in lae1 (CPK3793), wild type (QM 9414) and the LAE1-overexpressing tef1:lae1 strain CPK4086. Genes of interest are shown below regions of enrichment (y -axis scale, 030 for all tracks, all data were normalized for relative abundance). A. H3K4me3 is mildly enriched in the 5 region of cel1a but not nearly as strongly as in the neighbouring gene, which is not affected by changes in LAE1 expression. B. The most common pattern of CAZyme gene histone modifications tested here was no significant enrichment under any condition, represented here by cel1b . C. In contrast, highly expressed genes, like hH3 and hH4-1 show both H3K4me2 and -me3 but no H3K9me3 enrichment in a lae-1- independent manner. D. The same is true for a typical metabolic gene, spe1 , the gene for ornithine decarboxylase. E. The lae1 gene serves as control, as no signal is detected in lae1 , but H3K4me2 is enriched in the tef1:lae1 strain. F. A predicted gene encoding a protein with a carbohydrate-binding motif (protein ID 102735) shows enrichment of H3K4me2 only in the lae1 strain.

Figure 11: LAE1 affects sporulation in T. reesei . (A) Phenotype of T. reesei QM 9414 (WT), lae1 (CPK3793 on top, and CPK4086 below) and lae1 OE strains (CPK4086) on top, CPK3791 below); (B) effect of light on sporulation in T. reesei WT, lae1 (CPK3793) and lae1 OE (CPK4086) strains. Only one strain is shown, but consistent data have been obtained with at least two more strains of both lae1 genotypes.

Mentioned biomedical entities:

Genes: CA, CIP1, Cellulase, GH, GHs, HP1, HepA, KU70, LAE1, LaeA, MA, Tel, WI, XYN1, XYN2, XYR1, cbh2, cel12A, cel1A, cel3C, cel5A, cel6A, cel7A, cel7B, cellulase, cip1, cip2, hH4-1, histone, lae1, pki1, pyr4, tef1, xyn1, xyn2, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. fumigatus, A. nidulans, Escherichia coli, T. reesei

Tissues: N/A

Pathways: N/A

Query Result 5/25

Efficient genome editing in filamentous fungus Trichoderma reesei using the CRISPR/Cas9 system

Liu, Rui; Chen, Ling; Jiang, Yanping; Zhou, Zhihua; Zou, Gen

Cell Discovery

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Keywords Frequency (TF): cellulase: 0.001; reesei: 0.008; cbh1: 0.003; xyr1: 0.000

Abstract:

Filamentous fungi have wide applications in biotechnology. The CRISPR/Cas9 system is a powerful genome-editing method that facilitates genetic alterations of genomes in a variety of organisms. However, a genome-editing approach has not been reported in filamentous fungi. Here, we demonstrated the establishment of a CRISPR/Cas9 system in the filamentous fungus Trichoderma reesei by specific codon optimization and in vitro RNA transcription. It was shown that the CRISPR/Cas9 system was controllable and conditional through inducible Cas9 expression. This system generated site-specific mutations in target genes through efficient homologous recombination, even using short homology arms. This system also provided an applicable and promising approach to targeting multiple genes simultaneously. Our results illustrate that the CRISPR/Cas9 system is a powerful genome-manipulating tool for T. reesei and most likely for other filamentous fungal species, which may accelerate studies on functional genomics and strain improvement in these filamentous fungi.

Summary:

Introduction: Filamentous fungi have been used to produce diverse enzymes or metabolites for centuries. To obtain hyper-producers, repeated classical mutagenesis and screening have been applied for long-term strain improvements. Genetic engineering approaches for filamentous fungi have also been well developed. However, these approaches are not as efficient as those available for yeast and bacteria due to the additional complexity of filamentous filamentous. However , the time- and labor-intensive process of using zinc-finger nucleases and the requirements of specific enzyme engineering for different targets in transcription activator-like effector nucleolases limit the development of genome editing techniques for filamentously fungus.

Results: Expression of codon-optimized Cas9 in T. reesei. 2017 Kazusa. All rights reserved. - Authors: Kazushi. eGFP was fused to toCas9 and expressed in both the wild-type strain Qm6a and the mutant strain Rut-C30 under control of the Ppdc and Pcbh1 promoters. The expression of toC as9was expressed in each of the selected transformants.

Discussion: High-efficiency genetic manipulation approaches facilitate fungal strain improvement, as well as the elucidation of fungal molecular mechanisms behind the industrial applications. The first reports of genetic transformation in Aspergilluswere in 1983, which lagged behind those for Escherichia coliandS. cerevisiae. Over the past three decades, scientists have attempted to develop various tools or strategies to improve genetic manipulation strategies in filamentous fungi. Different genetic transformation approaches have been applied for homologous recombination in many filamentous yeasts; however, the low homologously recomination frequency (5) has limited studies on their functional genomics through targeted gene replacement.

Figures:

Figure 1: Expression and localization of Cas9-eGFP in T. reesei strains. (a) Detection of Cas9-eGFP expression in transformants 6a-pe and C30-ce by western blots. Transformants 6a-gfp and C30-gfp expressing eGFP were used as controls. Lanes C30-ce and 6a-pe were loaded with 0.05mg protein. Lanes C30-gfp and 6a-gfp were loaded only with 0.01mg protein to avoid overexposure. Anti-GFP antibody (Genscript, Nanjing, China) was used for detection. (b) Fluorescence microscopic assessment of the localization of Cas9-eGFP in C30-ce. The bars were 5m. DAPI, 4,6-diamidino-2-phenylindole; eGFP, enhanced green fluorescent protein.

Figure 2: CRISPR/Cas9-based mutagenesis and homologous recombination in T. reesei . (a) Alignments of ura5 mutants from 5-FOA-resistant colonies constitutive expression or post-lactose induction (the gene encoding Cas9 under control of Pcbh1 can be induced by oligosaccharides such as lactose, sophorose, or cellodextrin, in C30-cc) of Cas9 in gRNA transformants (14 6a-pc transformants and 15 C30-cc transformants). Tuar5 is the wild-type reference ura5 gene from T. reesei Qm6a and Rut-C30 (the PAM sequence is highlighted in blue and the guide RNA (gRNA)-guiding sequence in green). Deletions or insertions were found near the PAM sequence. (b) Schematic for homologous recombination (HR) of lae1, vib1, and clr2 mediated by Cas9 and donor DNAs. Results of PCR analyses revealed HR events.

Figure 3: Reduction of sporulation and cellulolytic capability in lae1 strains. (a) The observed sporulation of mutants and controls in potato dextrose agar plates after 7 days of growth. Sporulation was reduced in all three randomly selected lae1 strains compared with controls. (b) Assays for filter paper activity of lae1 strains and controls in inducing medium with 1% lactose after 3 days of fermentation. The cellulolytic capability was weakened significantly (P 0.01, t -test). Error bars represent the s.d. among three independent measurements.

Mentioned biomedical entities:

Genes: CA, Cas, Cas9, Ku, Ku70, Ku80, NEB, PAM, Ppdc, SNR52, U6, VIB1, cbh1, lae1, pdc, vib1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: E. coli, Escherichia coli

Tissues: mycelium

Pathways: N/A

Query Result 6/25

Regulation of Cellulase and Hemicellulase Gene Expression in Fungi

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Current Genomics

Year: 2013 PMID: 24294104 Citations: 86 Score: 146.22 DOI: 10.2174/1389202911314040002

Keywords Hits: cellulase: 180, reesei: 50, cbh1: 15, xyr1: 30

Keywords Frequency (TF): cellulase: 0.019; reesei: 0.005; cbh1: 0.002; xyr1: 0.003

Abstract:

Research on regulation of cellulases and hemicellulases gene expression may be very useful for increasing the production of these enzymes in their native producers. Mechanisms of gene regulation of cellulase and hemicellulase expression in filamentous fungi have been studied, mainly in Aspergillus and Trichoderma. The production of these extracellular enzymes is an energy-consuming process, so the enzymes are produced only under conditions in which the fungus needs to use plant polymers as an energy and carbon source. Moreover, production of many of these enzymes is coordinately regulated, and induced in the presence of the substrate polymers. In addition to induction by mono- and oligo-saccharides, genes encoding hydrolytic enzymes involved in plant cell wall deconstruction in filamentous fungi can be repressed during growth in the presence of easily metabolizable carbon sources, such as glucose. Carbon catabolite repression is an important mechanism to repress the production of plant cell wall degrading enzymes during growth on preferred carbon sources. This manuscript reviews the recent advancements in elucidation of molecular mechanisms responsible for regulation of expression of cellulase and hemicellulase genes in fungi.

Summary:

INTRODUCTION: Cellulase enzymes are required for the hydrolysis of polysaccharides in lignocellulosic biomass. The enzymes involved in the hydrolase of polymers are: -glucosidase, -1,4-xylanases, xylosidasses, endo--1-1, 4-xylases and -arabinofuranosidas. The enzymatic activities involved in cellulose hydrolysis are:

CONCLUSIONS: The regulation of (hemi)cellulolytic genes appears to be basically the same among filamentous fungi such as T. reesei, N. crassa, Asperigillusspp., although their regulatory mechanisms are quite complex and present some differences. In particular, cross talks between expression of cellulolysis and hemicellulase genes make the regulatory mechanisms more complicated. Xyr homologs (Xy1 ofT. reaesi, XlR ofN. aspergillus, and XnR ofAspergillusspp) mediate expression of both xylanolytic and cellulalytic genes in response to xilan.

Figures:

Figure 1: Schematic representation of transcriptional factors affecting cellulases and xylanases expression in T. reseei (black box), N. crassa (grey box) and Aspergillus spp. (white box). The carbon catabolite repressor CRE, the activators clbR, Xyr/xlR/XlnR,Clr,ACE2, the repressorACE1, the CCAAT binding Hap2/3/5 complex, the pH regulator PacC, and the nitrogen regulators AreA and Nit2 are shown. The repressionactivity ( ), the induction activity ( ) and also promoter ( ) and coding region ( ) are indicated.

Mentioned biomedical entities:

Genes: ACE1, ACE2, Ace1, Ace2, AceI, AreA, CBH, CEL3A, CEL5A, CEL6A, CEL7A, CLR-1, CPC1, CRE, CRE-1, CRE1, Cellulase, ClbR, CreA, CreB, CreD, GATA-1, HAP2, HAP3, HAP5, Hap, LAE1, LIM1, NF-YA, NFYB, NIT2, Nit2, PY, PacC, RUT, SNF1, XYR1, XlnR, Xyr1, abfB, ace1, ace2, areA, cbh1, cbh2, cel6a, cellulase, clbR, cmc1, cre-1, cre1, cre2, creA, egl1, gh10-2, gsn, hap2, histone, nag1, pelA, pmeA, swo1, xlnD, xlnR, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. aculeatus, A. fumigatus, A. nidulans, A. phoenicis, A. terreus, human

Tissues: N/A

Pathways: N/A

Query Result 7/25

Factors regulating cellulolytic gene expression in filamentous fungi: an overview

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Microbial Cell Factories

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Abstract:

The growing demand for biofuels such as bioethanol has led to the need for identifying alternative feedstock instead of conventional substrates like molasses, etc. Lignocellulosic biomass is a relatively inexpensive feedstock that is available in abundance, however, its conversion to bioethanol involves a multistep process with different unit operations such as size reduction, pretreatment, saccharification, fermentation, distillation, etc. The saccharification or enzymatic hydrolysis of cellulose to glucose involves a complex family of enzymes called cellulases that are usually fungal in origin. Cellulose hydrolysis requires the synergistic action of several classes of enzymes, and achieving the optimum secretion of these simultaneously remains a challenge. The expression of fungal cellulases is controlled by an intricate network of transcription factors and sugar transporters. Several genetic engineering efforts have been undertaken to modulate the expression of cellulolytic genes, as well as their regulators. This review, therefore, focuses on the molecular mechanism of action of these transcription factors and their effect on the expression of cellulases and hemicellulases.

Summary:

Introduction: The gradual depletion of conventional fossil fuels and increasing awareness about the effects of greenhouse gas emissions has led to global interest in the development of renewable energy, particularly biofuels such as ethanol. Biofuels can be categorized into 1st, 2nd, 3rd, and 4th generation biofuel based on the feedstock used for production. The 1sth generation Biofuel are produced by the fermentation of sugar or starchy substrates and these processes have already been commercially established. However, considering the huge demand for ethanol, it has become necessary to expand the scope of fermentable substrates beyond the conventional ones (e.g. cane molasses, starch, etc.) to lignocellulosic biomass, which is available in abundance.

Conclusion: This review provides an overview of the complex regulatory network involved in the expression of cellulase and hemicellulases genes in filamentous fungi. It provides insights into the regulatory factors involved in this process and provides more targets for the rational engineering of filamentous filamentous fungal strains to develop better strains, for the production of cellulose and hemicellulose enzymes. The on-site production of cellsulased enzymes seems to be the most feasible solution to overcome the challenge of using expensive commercial cellulolytic enzymes for the hydrolysis of biomass, in 2G ethanol biorefineries. However, filamentous plants like T. reesei, P. funiculosum, Ps. oxalicum etc. that can produce cellullases and hemiscellullases have several inherent limitations in their enzyme production and secretion capacity. These can be overcome by genetic engineering to develop strains that produce enzymes at an industrially relevant scale.

Figures:

Figure 1: Transcriptional regulators and transporters involved in cellulase expression in T. reesei and Penicillium sp

Figure 2: Comparative analysis of genome with respect to CAZymes in Trichoderma reesei , Penicillium funiculosum and Penicillium oxalicum 1142. GHs Glycoside hydrolases, GTs Glycosyl transferase, PLs Polysaccharide Lyases, CEs carbohydrate esterases, AAS auxiliary activities, and CBMs carbohydrate binding modules

Mentioned biomedical entities:

Genes: 586, Ace1, Ace2, Ace3, AceI, Are1, AreA, Azf1, B, BglR, C, CYC8, Cdt1, CdtC, Clr1, Clr2, ClrB, Cre1, Cre2, CreA, CreB, CreC, Crt1, Crz1, Ctf1, GH, GHs, Gal11, HapB, LaeA, MFS, Pac1, PacC, PalA, STP1, Stp1, TUP1, TacA, Vel2, Vel3, VelB, Vib1, XlnR, Xyr1, ada, amdS, bgl1, bgl2, cbh1, cbh2, cellulase, creA, eg1, egl1, egl3, hydrolase, pac1, pdc, tubB, xlnR, xyn1, xyn2, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: A. nidulans, Lantana, Pseudotsuga, T. reesei

Tissues: N/A

Pathways: N/A

Query Result 8/25

The mating type locus protein MAT1-2-1 of Trichoderma reesei interacts with Xyr1 and regulates cellulase gene expression in response to light

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Scientific Reports

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Abstract:
Cellulase production in the model cellulolytic fungus Trichoderma reesei is subject to a variety of environmental and physiological conditions involving an intricate regulatory network with multiple transcription factors. Here, we identified the mating type locus protein MAT1-2-1 as an interacting partner for the key transcriptional activator Xyr1 of T. reesei cellulase genes. Yeast two-hybrid and GST pulldown analyses revealed that MAT1-2-1 directly interacted with the putative transcription activation domain (AD, 767940 aa) and the middle homology region (MHR2, 314632 aa) of Xyr1. Disruption of the mat1-2-1 gene compromised the induced expression of cellulase genes with Avicel in response to light or with lactose. Chromatin immunoprecipitation (ChIP) demonstrated that MAT1-2-1 was recruited to the cbh1 (cellobiohydrolase 1-encoding) gene promoter in a Xyr1-dependent manner. These results strongly support an important role of MAT1-2-1 as a physiological cofactor of Xyr1, and suggest that MAT1-2-1 represents another regulatory node that integrates the light response with carbon source signaling to fine tune cellulase gene transcription.

Summary:

Introduction: The ascomycete Trichoderma reesei (teleomorph Hypocrea jecorna), a saprophytic filamentous fungus, represents an important workhorse for industrial production of cellulases as well as other proteins. The secreted cellulosic mixture synergistically decomposes the insoluble cellulusic materials to fermentable sugars. InT. reseei, most of the cellullase genes are coordinately controlled by a suite of transcription factors.

Results: Mating type locus (MAT1-2) is a transcription factor that interacts with Xyr1 in a yeast two-hybrid screen. XYR1 is enriched in the Y2H assay. YYS-XYr1AD and -XyR1MHR2 were effective in resolving the interaction. 2016 Elsevier B.V. All rights reserved. GB/T 7714

Discussion: MAT1-2-1 ofT. reesei interacts with Xyr1, binds to thecbh1promoter and modulates cellulase gene expression. The exact underlying mechanism is unclear. 2016 Elsevier B.V. All rights reserved. - 2015 Elsvier B, BV. All Rights Reserved. MAT1-1 of T. rheei is a mating type locus protein that interacts directly with xyr1 and bind to the cbH1promotor.

Figures:

Figure 1: MAT1-2-1 interacts with Xyr1 in vitro . (A) Yeast two-hybrid analyses of the interaction between MAT1-2-1 and different regions of Xyr1. Serial dilutions of yeast transformant cells harboring the indicated plasmids were spotted on double dropout medium (DDO, SD/Leu/Trp) and quadruple dropout medium (QDO, SD/Ade/His/Leu/Trp) containing AbA plates, respectively, and were allowed grow at 30C for 3 days. Transformant containing p53/T pair was used as the positive control. (B) MAT1-2-1 interacts with Xyr1 AD and Xyr1 MHR2 in an in vitro GST pull-down assay. Recombinant 6His-MAT1-2-1 was incubated with glutathione sepharose 4B bead-coupled GST-Xyr1 AD, GST-Xyr1 MHR2 or GST as a control. MAT1-2-1 distributed in different fractions of the assay was detected by western blot with anti-His antibody.

Figure 2: MAT1-2-1 is co-localized with Xyr1 in the nucleus. Germinated hypha of T. reesei transformant simultaneously expressing EGFP-MAT1-2-1 and mCherry-Xyr1 were analyzed for the respective fluorescence with a Nikon Eclipse 80i fluorescence microscope. The result shown represents one of at least two independent experiments.

Figure 3: Transcription of mat1-2-1 under different cultivation conditions. qRT-PCR analysis of the relative transcription level of mat1-2-1 . The mean values of each column were obtained from three independent biological replicates. Error bars indicated the standard error (SE) from three independent biological replicates. Significant differences (T-test P0.05, P0.01, P0.001) were detected for the transcription of mat1-2-1 when comparing light and darkness for the indicated inducing carbon sources.

Figure 4: Growth assay of the TU6-RP and mat1-2-1 strains on solid agar plate or in liquid MA medium under constant light (LL) or darkness (DD) condition. (A and B) Growth of the mat1-2-1 and TU6-RP strains on MM agar plates with the indicated carbon sources under constant light and darkness conditions, respectively. (C and D) The diameter of the colony measured after three days of growth. (E and F) Dry weight of the mat1-2-1 and TU6-RP strains grown in liquid MA medium with 1% glucose (w/v) at the indicated time points in constant light/darkness or in daylight with normal light-dark cycle.

Figure 5: Deletion of mat1-2-1 leads to down-regulation of cellulase gene expression on cellulose in light but not in darkness. (A and B) p NPC and p NPG hydrolytic activities of the extracellular supernatant of the mat1-2-1 and TU6-RP strains. (C to E) Relative expression levels of cbh1 (C), eg1 (D) and xyr1 (E) in these two strains cultured under the same conditions for enzymatic activity analyses were assayed by RT-qPCR. Error bars

indicated the standard error (SE) from three independent biological replicates. Significant differences (T-test P0.05, P0.01) were detected for the extracellular activities and the transcription of the cbh1 , egl1 and xyr1 genes between TU6-RP and the mat1-2-1 strain for the indicated time points after induction.

Figure 6: MAT1-2-1 plays an important role in lactose induced expression of cellulases. (A and B) p NPC and p NPG hydrolytic activities of the extracellular supernatant, and (C and D) the relative transcription levels of cbh1 and crt1 of the mat1-2-1 and TU-6-Rp strains cultured on lactose. Error bars indicated the SE from three independent biological replicates. Significant differences (T-test, P0.05, P0.01, P0.001) were detected for the extracellular p NPC and p NPG activities between TU6-RP and mat1-2-1 for the indicated time points after induction. Significant differences also detected for the transcription of the cbh1 and crt1 genes (T-test P0.05, P0.01).

Figure 7: MAT1-2-1 is recruited to the cbh1 promoter upon cellulase induction. ChIP analyses of MAT1-2-1 recruitment to the 800 and 200 regions of the cbh1 promoter (A and B), the pheromone receptor gene promoters (hpr1 and hpr2) (C), and the pheromone precursor gene promoters (hpp1 and pgg1) (D) before (Gly) and after induction with Avicel for the indicated time periods. (E) Semi-quantitative PCR products amplified from the above promoter fragments in the immunoprecipitated samples. GM and WT denoted samples from the EGFP-MAT1-2-1 and TU6-RP strains, respectively. Gly denoted pre-cultures on glycerol.

Figure 8: MAT1-2-1 recruitment to the cbh1 promoter is dependent on Xyr1. (A) Expression and subcellular localization of EGFP-MAT1-2-1 in the xyr1 strain. (B and C) ChIP analyses of MAT1-2-1 recruitment to the 800 and 200 regions of the cbh1 promoter in the presence or absence of Xyr1 after induction with Avicel for the indicated time periods. Gly denoted pre-cultures on glycerol. (D) ChIP analyses of MAT1-2-1 recruitment to different regions of the cbh1 promoter under Avicel inducing conditions in the EGFP-MAT1-2-1, xyr1 EGFP-MAT1-2-1 strains and a control strain expressing EGFP. Error bars indicated the SD from two independent biological replicates. The actin , hpr1 and hpr2 promoters were used as controls. Significant differences (T-test P0.05, P0.01, P0.001, n.s. not significant) were detected for the recruitment of EGFP-MAT1-2-1 to all the tested regions of the cbh1 promoter between the EGFP-MAT1-2-1 and the xyr1 EGFP-MAT1-2-1 strains. No significant recruitment was detected for MAT1-2-1 to the actin and hpr1 promoters or for GFP alone to all the tested promoters.

Figure 9: The putative regulatory effect of MAT1-2-1 on the cellulolytic response in T. reesei .

Mentioned biomedical entities:

Genes: AD, Ade2, B, ENV1, GST, HMG, His3, MA, MAT1-1, MAT1-2, MEL1, TF, VEL1, WT, Xyr1, actin, cbh1, cbh2, cellulase, crt1, eg1, egl1, env1, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: conidium

Organisms: E. coli, Escherichia coli, Escherichia coli BL21

Tissues: N/A

Pathways: N/A

Query Result 9/25

Improved Production of Majority Cellulases in Trichoderma reesei by Integration of cbh1 Gene From Chaetomium thermophilum

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Abstract:

Lignocellulose is an abundant waste resource and has been considered as a promising material for production of biofuels or other valuable bio-products. Currently, one of the major bottlenecks in the economic utilization of lignocellulosic materials is the cost-efficiency of converting lignocellulose into soluble sugars for fermentation. One way to address this problem is to seek superior lignocellulose degradation enzymes or further improve current production yields of lignocellulases. In the present study, the lignocellulose degradation capacity of a thermophilic fungus Chaetomium thermophilum was firstly evaluated and compared to that of the biotechnological workhorse Trichoderma reesei. The data demonstrated that compared to T. reesei, C. thermophilum displayed substantially higher cellulose-utilizing efficiency with relatively lower production of cellulases, indicating that better cellulases might exist in C. thermophilum. Comparison of the protein secretome between C. thermophilum and T. reesei showed that the secreted protein categories were quite different in these two species. In addition, to prove that cellulases in C. thermophilum had better enzymatic properties, the major cellulase cellobiohydrolase I (CBH1) from C. thermophilum and T. reesei were firstly characterized, respectively. The data showed that the specific activity of C. thermophilum CBH1 was about 4.5-fold higher than T. reesei CBH1 in a wide range of temperatures and pH. To explore whether increasing CBH1 activity in T. reesei could contribute to improving the overall cellulose-utilizing efficiency of T. reesei, T. reesei cbh1 gene was replaced with C. thermophilum cbh1 gene by integration of C. thermophilum cbh1 gene into T. reesei cbh1 gene locus. The data surprisingly showed that this gene replacement not only increased the cellobiohydrolase activities by around 4.1-fold, but also resulted in stronger induction of other cellulases genes, which caused the filter paper activities, Azo-CMC activities and -glucosidase activities increased by about 2.2, 1.9, and 2.3-fold, respectively. The study here not only provided new resources of superior cellulases genes and new strategy to improve the cellulase production in T. reesei, but also contribute to opening the path for fundamental research on C. thermophilum.

Summary:

Introduction: Plant biomass from agriculture and forestry is one of the most abundant resources on the earth and it has been considered as a promising material for producing renewable biofuels and other value-added bio-products. One of the bottlenecks of this process is the enzymatic degradation of biomass-derived polysaccharides. Currently, the filamentous fungi Trichoderma, Aspergillus, and Penicillium species are the most commonly used producer for lignocellulases in industry. However, the optimum temperature of enzymes from these species is normally from 30 to 50C at which the efficiency of biomass polysacharides saccharification is very low.

Results: Chaetomium thermophilumexhibited greater utilization efficiency of cellulose materials compared to that ofT. reesei. The results suggest that the cellulose degradation capacity ofC. thermophiliumwas evaluated and compared with that of T. reaesi. To alleviate potential effects caused by differential spore germination efficiency,C. sporulation was firstly grown in medium containing glucose as the sole carbon source and the same amount of wet mycelia (about 2.2 g) were then transferred into medium based on Avicel or sugarcane bagasse as the carbon source. The abundance of biomass accumulation, the extracellular protein concentration, and the filter paper activities in the supernatant were measured at different time points.

Discussion: This study demonstrates that a thermophilic fungus C thermophilumsecreted fewer cellulases, but showed higher efficiency to degrade celluloses. This finding indicated that C. thermophiliummight contain cellulolytic enzymes with higher specific activities. However, UPLC-MS/MS analysis of C.thermophilumsupernatants demonstrated there were many oxidoreductases included in C. thermalophilumprotein secretome.

Conclusion: To search for new resources of lignocellulases for plant biomass degradation, here we reported that a thermophilic fungus C. thermophilum could degrade celluloses more efficiently with less secreted cellulase compared to T. reesei, implying the existence of excellent cellulolytic genes in its genome. Comparison of the enzyme properties of CBH1 from C. tropphilum and T.reespi further confirmed this hypothesis. More interesting, our data showed that only raising the function of C. reaesi could lead to a marked increase in the production levels of other celllulases. Based on the observed phenomenon, we speculated that if celluloids are accumulated temporarily, they might play an important role in the process of celluleases genes induction.

Figures:

Figure 1: Phenotypic analysis of C. thermophilum and T. reesei when response to cellulose. (A) The intracellular protein concentration of C. thermophilum and T. reesei mycelia collected from the culture under Avicel or sugarcane bagasse after transferring from the glucose culture. (B) Growth phenotype of C. thermophilum and T. reesei under Avicel or sugarcane bagasse. About 2.2 g wet mycelia of C. thermophilum and T. reesei

were collected and transferred into Avicel or sugarcane bagasse media from glucose contained media and cultivated for 48 h. (C) Total secreted protein levels in the supernatants of C. thermophilum and T. reesei in Avicel or sugarcane bagasse media at different time points. (D) Filter paper activities of the supernatants of C. thermophilum and T. reesei in Avicel or sugarcane bagasse media at different time points. Shown in (A , C,D) are the mean values of three biological replicates. Error bars show the standard deviations between these replicates. T. reesei -A, T. reesei-S , C. thermophilum -A, and C. thermophilum -S respectively indicates T. reesei grown on Avicel (A) or sugarcane bassage (S) and C. thermophilum grown on Avicel (A) or sugarcane bassage (S). Shown in (B) is one of the three represents.

Figure 2: The extracellular protein categories in the supernatants of cellulose cultures were different between C. thermophilum and T. reesei. (A) SDS-PAGE analysis of the secreted proteins in the supernatant of C. thermophilum and T. reesei in 120-h Avicel cultures after transferring from glucose cultures. (B) Major protein categories of the secretome of C. thermophilum and T. reesei under Avicel. The data was collected by UPLC-MS/MS analysis and the detailed information was listed in Supplementary Tables S2 S7 .

Figure 3: Expression of Tr-CBH1 and Ct-CBH1 in T. reesei using cellulase-free background system. (A) Schematic diagram of the construction of recombinant strain Tr-cT cbh1 . (B) Schematic diagram of the construction of recombinant strain Tr-cC cbh1 . FgDNA-P indicates fragments in the parental strain. FvDNA indicates Fragment from plasmid vector constructed for the generation of the indicated recombinant strain. FgDNA-T indicates DNA fragments in the generated transformant strain.

Figure 4: Ct-CBH1 showed higher specific activity compared to Tr-CBH1. (A) SDS-PAGE analysis of the secreted proteins in the supernatant of the recombinant T. reesei strains C pyr4 , Tr-cT cbh1 , and Tr-cC cbh1 . A suspension of 10 6 conidia/mL was inoculated into MM media with glucose as the only carbon source and cultivated for 48 h. A total of 20 L of the supernatant was used for SDS-PAGE analysis. (B) Total secreted protein levels and CBH activities in the supernatants of the indicated T. reesei strains. The samples used in (B) were as same as in (A) . Shown are the mean values of three biological replicates. Error bars show the standard deviations between these replicates. The significance of differences between Tr-cT cbh1 and Tr-cC cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 5: Ct-CBH1 showed higher optimum temperature and better temperature stability compared to Tr-CBH1. (A) Optimum pH determination of Tr-CBH1 and Ct-CBH1. (B) pH stability analysis of Tr-CBH1 and Ct-CBH1. (C) Optimum temperature determination of Tr-CBH1 and Ct-CBH1. (D) Temperature stability analysis of Tr-CBH1 and Ct-CBH1. Shown are the mean values of three biological replicates. Error bars show the standard deviations between these replicates.

Figure 6: Replacement of Tr- cbh1 with Ct- cbh1 gene in T. reesei increased the production of cellulases. (A) Schematic diagram of the construction of recombinant T. reesei strain Tr-Ct cbh1 . FgDNA-P indicates the fragment in the parental strain. FvDNA indicates the fragment from plasmid vector constructed for generation of the indicated recombinant strain. FgDNA-T indicates DNA fragment in the generated transformant strain. (B) Total secreted protein levels of three independent C pyr4 and Tr-Ct cbh1 transformants in the supernatant of 120-h Avicel cultures after transferring from glucose cultures. Shown are the mean values of three technical replicates. Error bars show the standard deviations between these replicates. (C) SDS-PAGE analysis of the extracellular proteins of C pyr4 and Tr-Ct cbh1 strains. The protein bands highlighted with the red frame line indicates proteins in Tr-Ct cbh1 strains were less than that in C pyr4 strains. (D) Filter paper activities and CBH activities of C pyr4 and Tr-Ct cbh1 strains. The samples used in (C,D) were as same as in (B) . Error bars in (D) show the standard deviations between the biological replicates shown in (C) . The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 7: The activities of other cellulases were also increased in Tr-Ct cbh1 strains. (A) HPAEC-PAD analysis of glucose and cellobiose content in the solution of filter paper activity assay (FPA). nC, nano counts, indicating the intensity (abundance) of the signal for each peak. Glucose and cellobiose indicate 100 M glucose and cellobiose standard sample, respectively. Tr-Ct cbh1 and C pyr4 indicate sample taken from the solution of FPA assay with the supernatants of C pyr4 and Tr-Ct cbh1 strains under Avicel cultures, respectively. (B) Measurements of the endoglucanase activities and -glucosidase activities of C pyr4 and Tr-C tcbh1 strains in the supernatant of 120-h MM+2% Avicel cultures after transferring from 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 8: Increased CBH1 activities in T. reesei enhanced the induction and expression of the major cellulases genes. Relative quantification of mRNA levels of eg1 (A) , eg2 (B) , cbh2 (C) , and bgl1 (D) genes of C pyr4 and Tr-Ct cbh1 strains by qRT-PCR. RNA samples were obtained from the culture of MM+2% Avicel at the indicated time point after a shift of 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.05; P 0.001). ns, not significant.

Figure 9: The higher production of cellulases in Tr-Ct cbh1 strains was caused by increased expression level of xyr1. Relative quantification of mRNA levels of xyr1 (A) , cre1 (B) , ace1 (C) , ace2 (D) , and ace3 (E) genes of C pyr4 and Tr-Ct cbh1 strains by qRT-PCR. RNA samples were obtained from 12, 24, and 48 h of cultures of MM+2% Avicel after a shift of 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.05; P 0.01; P 0.001). ns,

not significant.

Figure 10: Graphic summary of this study. (A) Compared to T. reesei , C. thermophilum degrades celluloses more efficiently with lower production of cellulases. (B) The proposed model of the mechanism for higher production of cellulases in Tr-C tcbh1 strains. Replacement of Tr- cbh1 with Ct- cbh1 gene in T. reesei first increased the CBH1 activity, which could cause instantaneous accumulation of cellobiose. This extra cellobiose might function as a possible inducer to trigger more induction and expression of cellulases genes.

Mentioned biomedical entities:
Genes: CBH, CBH1, FPA, MA, MUC, Su, ace1, ace2, actin, bgl1, cbh1, cbh2, cel1b, cellulase, cre1, eg1, eg1, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: C. thermophilum

Tissues: N/A

Pathways: N/A

Query Result 10/25

Influence of cis Element Arrangement on Promoter Strength in Trichoderma reesei

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Keywords Frequency (TF): cellulase: 0.001; reesei: 0.002; cbh1: 0.022; xyr1: 0.002

Abstract:

Trichoderma reesei can produce up to 100 g/liter of extracellular proteins. The major and industrially relevant products are cellobiohydrolase I (CBHI) and the hemicellulase XYNI. The genes encoding both enzymes are transcriptionally activated by the regulatory protein Xyr1. The first 850 nucleotides of the cbh1 promoter contain 14 Xyr1-binding sites (XBS), and 8 XBS are present in the xyn1 promoter. Some of these XBS are arranged in tandem and others as inverted repeats. One such cis element, an inverted repeat, plays a crucial role in the inducibility of the xyn1 promoter. We investigated the impact of the properties of such cis elements by shuffling them by insertion, exchange, deletion, and rearrangement of cis elements in both the cbh1 and xyn1 promoter. A promoter-reporter assay using the Aspergillus niger, bgoxA gene allowed us to measure changes in the promoter strength and inducibility. Most strikingly, we found that an inverted repeat of XBS causes an important increase in cbh1 promoter strength and allows induction by xylan or wheat straw. Furthermore, evidence is provided that the distances of cis elements to the transcription start site have important influence on promoter activity. Our results suggest that the arrangement and distances of cis elements have large impacts on the strength of the cbh1 promoter, whereas the sheer number of XBS has only secondary importance. Ultimately, the biotechnologically important cbh1 promoter can be improved by cis element rearrangement.

Summary:

INTRODUCTION: Xyr1-binding sites (XBS) are a group of XBS that are found in the promoters of the cbh1and xyn1genes. XYR1 is a protein that is regulated by the XBP1 gene. Currently, the main research efforts focus on the transcriptional regulation of cellulase- and hemicellulases-encoding genes. Nonetheless, the impact of the arrangement ofciselements on promoter strength has hardly been investigated so far. Therefore, during this study, we investigated the role ofcisselements composed of xBS in the promors.

RESULTS: Theciselement IR increases promoter strength. We constructed expression cassettes bearing the deletion of the respectivecisselement in each promoter or a motif exchange where X was exchanged for IR in thecbh1promoter and termedcislelement X-IR and IR was exchange for X in thexyn1promotor and termedescribedciselelement Y-IR. The modified and native promoters were fused to theAspergillus nigergoxAgene, which is an established reporter system inTrichoderma species and subsequently transformed intoT. reeseiQM6atmus53pyr4. Using the reestablishment of pyr4as a marker, we investigated these strains by a carbon source replacement experiment.

DISCUSSION: In the past, relevant regions and motifs for the regulation of the xyn1 gene were identified. In this study, we present results demonstrating that the previously identified cis element IR, an inverted repeat of XBS, conveys transcriptional induction on xilan and wheat straw. We studied the role of this cs element by deleting only this specific motif and not by classic deletion analysis of the promoter. Phenotypic analysis was initially performed by carbon source replacement experiments because growth effects can be excluded and this allowed us to focus on the impact of a single carbon source on the promoteer activity.

Figures:

Figure 1: Schematic representation of the cbh1 and xyn1 promoters. Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the cbh1 or xyn1 gene. The blue box indicates the cis element Z (direct repeat of three XBS), the orange box indicates the cis element Y (an inverted repeat of XBS), the purple box indicates the cis element X (direct repeat of three XBS), and the brown box indicates the cis element IR (inverted repeat of XBS). Numbers on top indicate the distance from the ATG in base pairs.

Figure 2: Promoter schemes and GoxA activities of recombinant T. reesei strains after a carbon source replacement experiment. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were pregrown, and then mycelia were transferred to MA medium without any carbon source (gray bars) or with 1% glycerol (blue bars), 1% d -glucose (red bars), 0.5 mM d -xylose (purple bars), 1% d -xylose (green bars), or 1.5 mM sophorose (orange bars), incubated for 8 h, and harvested, and GoxA activities were measured in the supernatants. GoxA activity is given in relation to biomass (dry weight). Symbols indicate results below quantification level () and results below the detection limit (). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 3: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate

cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 4: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 5: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 6: GoxA activities of strains bearing combinations of modified cbh1 promoter fragments after direct cultivation. Thirty-five recombinant strains bearing different combinations of 3 and 5 fragments of the cbh1 promoter (compare Table 3) were grown on MA medium with 1% xylan (A), lactose (42 h cultivation time) (B), or pretreated wheat straw (C). The top value in each table cell gives the GoxA activity relative to that in strain PcWT. The bottom value in each table cell gives the absolute GoxA activity in relation to biomass, i.e., U/mg NaOH-soluble protein (A, C) or U/mg dry weight (B). Color key: the GoxA activity values of PcWT are highlighted in yellow; blue indicates lower and green indicates higher GoxA activities. The mean values shown are calculated from the results of technical triplicates. Error bars indicate standard deviations.

Mentioned biomedical entities:
Genes: AG, Ace1, Ace2, B, CA, Cre1, Gal4, GoxA, HRP, MA, MO, MP, TetR, WI, Xyr1, cbh1, cellulase, clb2, cyc, goxA, gpd, leu3, peroxidase, pyr4, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: Escherichia coli

Tissues: capillary

Pathways: N/A

Query Result 11/25

Trichoderma reesei XYR1 activates cellulase gene expression via interaction with the Mediator subunit TrGAL11 to recruit RNA polymerase II

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Keywords Frequency (TF): cellulase: 0.009; reesei: 0.004; cbh1: 0.003; xyr1: 0.013

Abstract:

The ascomycete Trichoderma reesei is a highly prolific cellulase producer. While XYR1 (Xylanase regulator 1) has been firmly established to be the master activator of cellulase gene expression in T. reesei, its precise transcriptional activation mechanism remains poorly understood. In the present study, TrGAL11, a component of the Mediator tail module, was identified as a putative interacting partner of XYR1. Deletion of Trgal11 markedly impaired the induced expression of most (hemi)cellulase genes, but not that of the major -glucosidase encoding genes. This differential involvement of TrGAL11 in the full induction of cellulase genes was reflected by the RNA polymerase II (Pol II) recruitment on their core promoters, indicating that TrGAL11 was required for the efficient transcriptional initiation of the majority of cellulase genes. In addition, we found that TrGAL11 recruitment to cellulase gene promoters largely occurred in an XYR1-dependent manner. Although xyr1 expression was significantly tuned down without TrGAL11, the binding of XYR1 to cellulase gene promoters did not entail TrGAL11. These results indicate that TrGAL11 represents a direct in vivo target of XYR1 and may play a critical role in contributing to Mediator and the following RNA Pol II recruitment to ensure the induced cellulase gene expression.

Summary:

Introduction: In eukaryotes, RNA polymerase II (Pol II) transcribes all protein-coding genes whose initiation is dependent on a set of general transcription factors (GTFs), which recognize the core promoter and facilitate transcription initiation from the correct start site. While various additional factors contribute to the regulation of Pol II activity at the promoter, the multisubunit Mediator complex has been shown to be critical for expression of most, if not all, Pol 2 transcripts.

Results: Identification of a T.reeseiGAL11/MED15 homolog that interacts with XYR1in vitro. (A) Schematic diagram of domain of TrGAL11. The KIX domain and glutamine rich region (Q-rich) of each protein are labeled as indicated. (B) Yeast two-hybrid analyses of interactions between Xyr1 activation domain (XYr1 AD) and full length or the KIx domain of TrGAL1in vivo.

Discussion: In this study, we showed that the major transactivator XYR1 directly targets the Mediator tail module subunit TrGAL11 to initiate (hemi)cellulase gene transcription in T. reesei. Tail module sub units including TrGal11 and TrMED5 were demonstrated to be differentially involved in the cellulasese gene expression. In addition, TrGaL11 and thus RNA Pol II binding to cellulasse gene promoters specifically relied on the expression of XyR1, whereas the steady state XYA binding to its target regulatory sequences was subject to a regulation imposed by TrGALA11 recruitment.

Figures:

Figure 1: TrGAL11 interacts with XYR1 in vitro .

Figure 2: Trgal11 disruption had little effect on T . reesei growth on plates or in liquid medium but reduced its conidia and pigment formation.

Figure 3: Deletion of Trgal11 compromised the fully induced production of cellobiohydrolase and endoglucanase but not of -glucosidase.

Figure 4: Deletion of Trgal11 resulted in a significant decrease in the transcription of cellobiohydrolase and endoglucanase but not -glucosidase genes.

Figure 5: Overexpression of xyr1 failed to restore the fully induced expression of cellulase genes in the Trgal11 deletion mutant.

Figure 6: TrGAL11 was recruited to cellulase gene promoters in an XYR1-dependent manner.

Figure 7: Trgal11 deletion resulted in a significantly higher XYR1 occupancy on cellulase gene promoters.

Figure 8: TrGAL11 plays a critical role in RNA Pol II recruitment on the core-promoter of cbh / eg genes but not of bgl genes.

Figure 9: A schematic model of how Mediator is recruited by XYR1 to participate in activating cellulase gene expression in T . reesei .

Mentioned biomedical entities:

Genes: B, Blue, CKM, FL, FPA, GAL4, GCN4, GST, HSF1, IP, MED14, MED16, MED2, MED3, MEL1, PIC, Rpb1, SD, TFIIH, UPS, XYR1, actin, bgl1, bgl2, cbh, cbh1, cbh2, cellulase, eg1, p53, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: E. coli, human

Tissues: Tail

Pathways: N/A

Query Result 12/25

Botrytis cinerea Transcription Factor BcXyr1 Regulates (Hemi-)Cellulase Production and Fungal Virulence

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mSystems

Year: 2022 PMID: 36468854 Citations: 8 Score: 109.88 DOI: 10.1128/msystems.01042-22

Keywords Hits: cellulase: 25, reesei: 5, cbh1: 0, xyr1: 161

Keywords Frequency (TF): cellulase: 0.004; reesei: 0.001; cbh1: 0.000; xyr1: 0.023

Abstract:

Botrytis cinerea is an agriculturally notorious plant-pathogenic fungus with a broad host range. During plant colonization, B. cinerea secretes a wide range of plant-cell-wall-degrading enzymes (PCWDEs) that help in macerating the plant tissue, but their role in pathogenicity has been unclear. Here, we report on the identification of a transcription factor, BcXyr1, that regulates the production of (hemi-)cellulases and is necessary for fungal virulence. Deletion of the bcxyr1 gene led to impaired spore germination and reduced fungal virulence and reactive oxygen species (ROS) production in planta. Secreted proteins collected from the bcxyr1 deletion strain displayed a weaker cell-death-inducing effect than the wild-type secretome when infiltrated to Nicotiana benthamiana leaves. Transcriptome sequencing (RNA-seq) analysis revealed 41 genes with reduced expression in the bcxyr1 mutant compared with those in the wild-type strain, of which half encode secreted proteins that are particularly enriched in carbohydrate-active enzyme (CAZyme)-encoding genes. Among them, we identified a novel putative expansin-like protein that was necessary for fungal virulence, supporting the involvement of BcXyr1 in the regulation of extracellular virulence factors.

Summary:

INTRODUCTION: Botrytis cinerea is a well-known plant-pathogenic fungus that is capable of infecting more than 1,000 plant species and is responsible for huge economic losses worldwide. During the infection process, B. cinereasecrets a wide range and large amounts of plant-cell-wall-degrading enzymes (PCWDEs) that macerate the plant tissue. Due to their high abundance and role in tissue degradation, PCWDE have been considered virulence factors and have been studied for close to a century). However, despite intensive studies, there is no direct evidence for a role in pathogenicity for the vast majority of theB. cineREAPCWDEAs).

RESULTS: BcXyr1 is required for (hemi-)cellulase production in B. cinerea. It is a transcription factor that regulates cellulose and xylanasasis. Its function is related to the transcription factor XylnR. The halo zones of the three strains (wild type, ox-bcxyr1, and obc-bxylr1) were similar.

DISCUSSION: Virulence of B. cinerea is mediated by the transcription factor BcXyr1 and the expansionin-like gene bcexl1 (bcxyr1), which is required for full virulence. The transcription factor Xylr1/XlnR is a major regulator of genes encoding xylanolytic and cellulolytic enzymes in different fungi. In this study, deletion of bbcyr1/bccyyr2reduced the secretion of xilanases and cellulose, confirming the regulation of (hemi-)cellulase genes by Bcxr1. This result was further confirmed by RNA-seq analysis, which revealed 41 genes that were positively regulated by BbcXr1 of which 13 genes were categorized as CAZymes. Deletion of a Bcyl1-regulated expansin- like gene (bbexl1) was required for complete virulentity.

Figures:

Figure 1: bcxyr1 is required for (hemi)cellulase production in B. cinerea . (a) Colonies were initiated by placing a 5-L droplet containing 500 spores of wild-type (wt), bcxyr1 , or ox- bcxyr1 strains onto a CMC plate, and the plates were incubated for 4 days and then stained by Congo red. (b) Average halo diameter of each strain after incubation for 4 days. (c) Three strains were grown in liquid Gamborg medium supplemented with 2% rice straw powder as a (hemi)cellulase inducer, and after 1 day of growth, 20L of culture supernatant of each strain was subjected to SDS-PAGE and then silver staining. (d) Xylanase activity and CMCase activity of culture supernatant samples of c were measured. Values that are statistically significantly different (P 0.01) by two-tailed Students t test from the wild-type values are indicated by an asterisk.

Figure 2: Deletion of bcxyr1 affects spore germination. (a) Spore germination in CD+1% sucrose medium. Spores were diluted to 10 5 /mL in liquid CD+1% sucrose medium. A total of 15L of spore suspension of each strain was pipetted onto glass coverslips and incubated for 6h. (b) Spore germination percentage of each sample after incubation for 6h was counted under an inverted microscope. Four replicates were used for each sample. Values that are statistically significantly different (P 0.01) by two-tailed Students t test from the wild-type values are indicated by an asterisk. (c) A total of 7.5L of spore suspension (10 5 /mL) in liquid CD+1% sucrose medium was inoculated onto Arabidopsis leaves, images were taken at 8 hpi, and the scale bar indicates 20m.

Figure 3: bcxyr1 is required for fungal virulence. (a) The culture supernatants (20g/mL protein in PBS) from strains which grew 24h in Gamborg medium supplemented with 2% rice straw powder were infiltrated into N. benthamiana leaves, and necrosis was observed 5days after infiltration. PBS was used as a mock control. The black circles indicate the infiltrated area. (b) Arabidopsis leaves were inoculated with 7.5-L droplets of spore suspension (10 5 spores/mL in liquid CD+1% sucrose medium), and typical lesions were developed at 80 hpi. (c) Average lesion size of sample b, and data are the means plus SD for eight lesions from three plants. (d) Arabidopsis leaves were inoculated with mycelial plugs (3mm in diameter) from a CD+1% sucrose agar plate, and typical lesions were developed at 34 hpi. (e) Average lesion size of sample d, and data are the means plus SD for

eight lesions from three plants. Values that are statistically significantly different (P 0.01) by two-tailed Students t test from the wild-type values are indicated by an asterisk.

Figure 4: Arabidopsis leaves were inoculated with 7.5-L droplets of spore suspension (10 5 spores/mL in liquid CD+1% sucrose medium). Leaves were cut at the designated time points, and samples were stained with lactophenol trypan blue. The large images show the entire lesion; squares show hyphal orientation and structure at the colony edge. Pictures were taken with the same magnification, and scale bar indicates 500m.

Figure 5: Analysis of differentially expressed genes in the bcxyr1 mutant compared the wild-type strain. (a) eggNOG function classification of downregulated genes (left) and upregulated genes (right) in the bcxyr1 strain. (b) KEGG pathways enrichment analysis of downregulated genes (left) and upregulated genes (right) in the bcxyr1 strain.

Figure 6: Deletion of bcexl1 affects fungal virulence. Spores were diluted to 10 5 /mL in liquid CD+1% sucrose medium, and 7.5L of spore suspension was inoculated onto Arabidopsis leaves. (a) Typical lesions developed at 75 hpi. (b) Average lesion size at 75 hpi. Data are the means SD of eight leaves for each strain. Columns marked by different letters represented statistical differences (P 0.01).

Figure 7: Double deletion of bcexl1 and bcxyr1 results in further reduction of fungal virulence. Spores were diluted to 10 5 /mL in liquid CD+1% sucrose medium, and 7.5L of spore suspension was inoculated onto Arabidopsis leaves. (a) Typical lesions developed at 84 hpi. (b) Average lesion size at 84 hpi. Data are the means SD of eight leaves for each strain. Columns marked by different letters represented statistical differences (P 0.01).

Mentioned biomedical entities:
Genes: Bcpg1, Bcpg2, CDIP, FIG, GHs, H2B, NEB, Tel, XlnR, Xyl1, Xyn11A, Xyr1, acr, bcpg2, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Tobacco, tomato

Tissues: tissue

Pathways: N/A

Query Result 13/25

Regulation of gene expression by the action of a fungal lncRNA on a transactivator

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RNA Biology

Year: 2020 PMID: 31517564 Citations: 6 Score: 101.72 DOI: 10.1080/15476286.2019.1663618

Keywords Hits: cellulase: 1, reesei: 16, cbh1: 1, xyr1: 287

Keywords Frequency (TF): cellulase: 0.000; reesei: 0.002; cbh1: 0.000; xyr1: 0.028

Abstract:

Long non-coding RNAs (lncRNAs) are crucial factors acting on regulatory processes in eukaryotes. Recently, for the first time in a filamentous fungus, the lncRNA HAX1 was characterized in the ascomycete Trichoderma reesei. In industry, this fungus is widely applied for the high-yield production of cellulases. The lncRNA HAX1 was reported to influence the expression of cellulase-encoding genes; interestingly, this effect is dependent on the presence of its most abundant length. Clearly, HAX1 acts in association with a set of well-described transcription factors to regulate gene expression. In this study, we attempted to elucidate the regulatory strategy of HAX1 and its interactions with the major transcriptional activator Xylanase regulator 1 (Xyr1). We demonstrated that HAX1 interferes with the negative feedback regulatory loop of Xyr1 in a sophisticated manner and thus ultimately has a positive effect on gene expression.

Summary:

Introduction: In recent decades, long non-coding RNAs (lncRNAs) have emerged as crucial players in regulatory processes in various eukaryotic organisms. Similar to other non-coded RNA species, lnnRNAs do not code for proteins but function directly as RNA. However, the group of lincRNA species is highly diverse and is distinguished from small functional ncRN species by their size (more than 200 nt), rather than by certain common characteristics ,.

Results: Fungal lncRNAHAX1and the transactivator Xyr1 can bind.As the recently discovered lincRNA HAX1, a physical interaction ofHAX1/XyR1 can be hypothesized. To test this possibility, RNA electrophoretic mobility shift assays were performed. To this end, the different versions ofHAx1varying in length (i.e.,HAX1262,HAX1299andHAX1428) were synthesizedin vitro, and XYr1 was expressed heterologously. The addition of increasing amounts of Xr1 toHAX1.

Discussion: In general, negative feedback regulation is a common strategy to ensure that a process stops when high amounts of the product are available. This regulation is essential to avoid unnecessary energetically demanding reactions and to maintain balance in complex metabolic pathways. However, despite the central role of Xyr1 in the regulation of PBDE expression, prior to this study, no evidence for possible auto-regulation of this transactivator had been reported. In this study we demonstrate that XYR1 regulates the expression of its own gene in a negative feedback loop via binding to the currently undescribed recognition site XRE based on bothin vitroandin vivoexperiments. Remarkably, the X RE was found to be strikingly different in terms of its properties as a XYA-binding motif compared to the YBS present in the promoters of xyr1.

Figures:

Figure 1: Analysis of the protein-RNA interaction of Xyr1 and HAX1 . RNA-EMSAs using either version of the lncRNA HAX1 , i.e., HAX1 262 (A), HAX1 299 (B) or HAX1 428 (C), and increasing amounts of Xyr1. An up to 4-fold molar excess of Xyr1 relative to 1g of in vitro synthesized HAX1 RNA was used.

Figure 2: Analyses of altered versions of the xyr1 promoter. (A) Schematic drawing of the xyr1 promoter (red bar). Previously described regulatory elements, such as Cre1-binding sites (yellow triangles), a Xyr1-binding site (green triangle) and a CAAT-box (blue box), as well as the newly identified 12 bp palindromic DNA motif XRE (pink box), are marked. The approximate positions of truncations are indicated by the numbers given on top (bp from ATG). (B) GoxA activities of recombinant T. reesei strains expressing the reporter gene goxA under the control of one of the truncated promoter versions (p804, p606, p497, p372) or the full-length version (pxyr1). The strains were pre-grown and replaced to minimal medium without a carbon source (blue bars) or to minimal medium with sophorose (orange bars). Means of biological replicates are derived from independently generated strains (n =2 for p804, p606, p497 and p372; n =3 for pxyr1). Error bars depict standard deviations; ANOVA followed by a post-hoc Tukey multiple comparison test (P 0.05) resulted in F (9;12)=11.684. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (C) GoxA activities of recombinant T. reesei strains cultivated as described above and expressing the reporter gene goxA under the control of the full-length promoter (pxyr1) or the full-length promoter lacking XRE (pXRE). Means of biological replicates are derived from independently generated strains (n =2 for pXRE; n =3 for pxyr1). Error bars depict standard deviations; t-tests resulted in t (3)=4.661, p 0.01 for the cultivation without carbon source and t (3)=3.326, p 0.05 for the cultivation on sophorose. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3.

Figure 3: Analysis of the binding of Xyr1 to XRE. (A) EMSA using 33.4 ng of the FAM-labelled probe containing the XRE alone (lane 1) or together with a 0.5-fold, 2-fold or 8-fold molar excess of Xyr1 relative to the probe (lanes 24). (B) Control EMSA using 33.4 ng of the FAM-labelled probe alone (lane

1) or with an 8-fold molar excess of Xyr1 (lane 2) or with an 8-fold molar excess of Xyr1 and increasing amounts (0.1-fold, 0.5-fold and 4-fold molar excess) of the unlabelled probe (Comp) (lanes 35) or with an 8-fold molar excess of BSA (lane 6).

Figure 4: Comparative analysis of the binding of Xyr1 to XRE and XBS. EMSAs using the FAM-labelled probe containing the XRE (Probe XRE) or the FAM-labelled probe containing XBS (Probe XBS) together with Xyr1, the unlabelled probe containing XBS (Comp XBS) or the unlabelled probe containing the XRE (Comp XRE). A total of 33.4 ng of the FAM-labelled probe alone (lanes 1) or together with a 4-fold molar excess of Xyr1 (lanes 2) or together with a 4-fold molar excess of Xyr1 and equimolar amounts of either of the unlabelled probes (lanes 3 and 4) were applied.

Figure 5: Analyses of the binding of Xyr1 and HAX1 to XRE and XBS. EMSAs using the FAM-labelled probe containing the XRE (A) or the FAM-labelled probe containing XBS (B) alone (lanes 1) or together with Xyr1 (lanes 2), or together with in vitro -synthesized CY5-labelled HAX1 428 (HAX1) (lanes 4) or together with both (lanes 3). As a control, HAX1 428 was also loaded alone (lanes 5). An 8-fold molar excess of Xyr1 and a 4-fold molar excess of RNA relative to 33.4 ng of the FAM-labelled probes were used. Fluorescence and image analyses were performed via FAM-scan (right images), CY5-scan (middle images) and ethidium bromide staining (left images).

Figure 6: Analyses of the binding of Xyr1 to XRE and XBS in the presence of HAX1 . EMSAs using the FAM-labelled probe containing the XRE (A) or the FAM-labelled probe containing XBS (B) alone (lanes 1) or together with different amounts of Xyr1 (lanes 24), or together with 8-fold molar excess of Xyr1 and increasing amounts of HAX1 262 (lanes 57) or together with 8-fold molar excess of Xyr1 and increasing amounts of HAX1 428 (lanes 810). (C) RNA-EMSA using 0.5g of in vitro -synthesized CY5-labelled HAX1 299 (HAX1 -CY5) alone (lane 1), or together with an 8-fold molar excess of Xyr1 (lane 2), or together with an 8-fold molar excess of Xyr1 and equimolar amounts of the FAM-labelled probe containing XBS (lane 3) or together with an 8-fold molar excess of Xyr1 and a 4-fold molar excess of the FAM-labelled probe containing XBS (lane 4). Fluorescence and image analyses were performed via FAM-scan (right image), CY5-scan (middle image) and ethidium bromide staining (left image).

Figure 7: Analyses of the impact of HAX1 on xyr1 expression. (A) Transcript levels of xyr1 in the presence and absence of hax1 . The hax1 deletion strain QM6a hax1 (green bars) and the parent strain QM6aloxP (blue bars) were pre-grown and transferred to minimal medium without carbon source or containing xylose (XO) or sophorose (S) for 3h. Transcript analysis was performed in technical triplicates. Transcript levels were normalized to act and sar1 , refer to cultivation without carbon source and are given in logarithmic scale (lg). Error bars indicate standard deviations; t-tests resulted in t (4)=7.948, p 0.01 for the cultivation on xylose and t (4)=4.395, p 0.05 for the cultivation on sophorose. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (B) Transcript levels of xyr1 in the presence or absence of hax1 824 overexpressed in trans . The hax1 overexpression strain OE hax1 (purple bars) and the reference strain QM6a tmus53 (blue bars) were pre-grown and transferred to minimal medium containing sophorose for 1h and 2h. Transcript analysis was performed in technical triplicates. Transcript levels were normalized to act and sar1 , refer to QM6a tmus53 cultivated for 1h and are given in logarithmic scale (lg). Error bars indicate standard deviations; t-tests resulted in t (4)=1.672, p 0.05 for the cultivation for 1h and t (4)=4.380, p 0.05 for the cultivation for 2h. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (C) GoxA activities of strains expressing the reporter gene goxA under the control of the full-length xyr1 promoter (pxyr1, blue bar) or the full-length xyr1 promoter lacking XRE (pXRE, orange bar) and a strain expressing the reporter gene goxA under the control of the full-length xyr1 promoter and overexpressing hax1 428 (pxyr1OE hax1 , red bar). The strains were pre-grown and transferred to minimal medium containing sophorose for 2h. GoxA assays were performed in technical triplicates. Error bars indicate standard deviations; t-tests resulted in t (4)=3.239, p 0.05 for the analysis of pxyr1OE hax1 versus pxyr1 and t (4)=0.292, p 0.05 for the analysis of pxyr1OE hax1 versus pXRE. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3.

Figure 8: Schematic illustration of the proposed regulatory model of HAX1 and Xyr1. Xyr1 (blue hexagon) can bind to two DNA motifs: XBS and XRE. Preferentially, Xyr1 binds to XBS at the promoter regions of its target genes (e.g., cbh1, cbh2, xyn1), thereby acting as a transcriptional activator (left section). When high amounts of Xyr1 are also available, the XRE on the xyr1 promoter gets bound, thereby resulting in a negative feedback regulation of xyr1 expression (right section). Via formation of an RNA-protein complex, HAX1 interferes with the negative feedback regulatory loop of Xyr1, rather than with its function as an activator. Thus, HAX1 finally has an enhancing effect on the expression of the Xyr1 target genes. Due to a higher affinity of Xyr1 for the longer versions of HAX1, HAX1 428 has the largest impact on PBDE expression, and HAX1 299 has a greater effect than the shortest version, HAX1 262.

Mentioned biomedical entities:
Genes: Cre1, GoxA, HAX1, MA, MO, MP, NEB, WI, XRE, Xyr1, bgl1, cbh1, cbh2, goxA, pyr4, sar1, tef1, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell

Organisms: Escherichia coli

Tissues: pulp

Pathways: N/A

Query Result 14/25

A homologous production system for Trichoderma reesei secreted proteins in a cellulase-free background

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Applied Microbiology and Biotechnology

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Keywords Frequency (TF): cellulase: 0.019; reesei: 0.006; cbh1: 0.002; xyr1: 0.007

Abstract:

Recent demands for the production of biofuels from lignocellulose led to an increased interest in engineered cellulases from Trichoderma reesei or other fungal sources. While the methods to generate such mutant cellulases on DNA level are straightforward, there is often a bottleneck in their production since a correct posttranslational processing of these enzymes is needed to obtain highly active enzymes. Their production and subsequent enzymatic analysis in the homologous host T. reesei is, however, often disturbed by the concomitant production of other endogenous cellulases. As a useful alternative, we tested the production of cellulases in T. reesei in a genetic background where cellulase formation has been impaired by deletion of the major cellulase transcriptional activator gene xyr1. Three cellulase genes (cel7a, cel7b, and cel12a) were expressed under the promoter regions of the two highly expressed genes tef1 (encoding translation elongation factor 1-alpha) or cdna1 (encoding the hypothetical protein Trire2:110879). When cultivated on d-glucose as carbon source, the xyr1 strain secreted all three cellulases into the medium. Related to the introduced gene copy number, the cdna1 promoter appeared to be superior to the tef1 promoter. No signs of proteolysis were detected, and the individual cellulases could be assayed over a background essentially free of other cellulases. Hence this system can be used as a vehicle for rapid and high-throughput testing of cellulase muteins in a homologous background.

Summary:

Introduction: Fungal cellulases are attractive for enzymatic cellulose conversion as they are highly active and can be expressed at levels exceeding 100g per liter in fungal hosts such as Trichoderma reesei. Lignocellulose, composed mainly of cellulose, hemicellulose, and lignin, is the most abundant renewable carbon source on earth and therefore an attractive resource to use it for the production of different chemical building blocks or biofuels.

Results: Cellulase expression vector construction for the expression of cellulases in a xyr1strain inT. reesei. The expression of three cellullases was tested under both promoters. The cellullases were amplified by PCR and ligated downstream of thetef1promoter region. The cdna1 gene was cloned from the cDNA1 gene.

Discussion: XYR1 transcriptional regulator of cellulase and hemicellulases was not feasible, we show that the cloning of an expression vector based on the cdna1expression signals result in a reasonable high production of these enzymes ond-glucose as carbon source. In contrast to earlier reports where the coding sequence of the expression vector was not available, we present a novel approach to the characterization of cellulose degradation enzymes in T. reesei.

Figures:

Figure 1: Expression plasmids for cellulase production. Schematic presentation of the two cellulase expression plasmids used in this study. Both plasmids contain the hygromycin B expression cassette as fungal selection marker followed by either the tef1 or the cDNA1 promoter region and the coding and terminator region of the respective cellulase gene (e.g., the endoglucanase encoding gene cel7b)

Figure 2: Endoglucanase CEL7B, endoglucanase CEL12A, and cellobiohydrolase CEL7A production in the cellulase negative background of T. reesei xyr1 transformants. Expression profile of typical xyr1 transformants expressing CEL7B (a), CEL12A (b), or CEL7A (c) under the tef1 (t) or cdna1 (c) promoter regions as indicated. Left SDS-PAGE gel of the supernatant of selected transformants after Coomassie blue staining. The amount of supernatant loaded to each gel is indicated in parentheses. Right total extracellular protein concentration (in milligrams per liter) of selected transformants expressing the respective cellulase under the cDNA1 (filled diamond) or tef1 promoter region (filled square). xyr1 (empty triangle) is included as a negative control

Figure 3: Cellulase activities in the supernatants of the cel7b , cel7a , and cel12a transformants hydrolysis rates of MUC, MULAC, and CMC by culture supernatants of CEL7B (a), CEL12A (b), and CEL7A (c) producing strains respectively were assayed at each sampling point. Volumetric activities (dashed lines) are reflective of total active enzyme amount in the culture supernatant and specific activities (solid lines) are reflective of enzyme purity. Compared to the transformants the parental strain T. reesei xyr1 (empty triangle) has negligible activities towards MUC, MULAC, and CMC. The strains selected are identical to Fig. 2 and express the respective cellulase either under the cDNA1 (filled diamond) or tef1 promoter region (filled square). In the case of strain c-cel7b-6, extra d -glucose (10g/L) was added after 20h of incubation (empty diamond)

Mentioned biomedical entities:

Genes: AG, CEL12A, CEL3A, CEL6A, CEL7A, CEL7B, Cellulase, MUC, Tef1, XYR1, cbh1, cel12a, cellulase, egl3, tef1, xlnR, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 15/25

Inducer-free cellulase production system based on the constitutive expression of mutated XYR1 and ACE3 in the industrial fungus Trichoderma, breesei

Arai, Toshiharu; Ichinose, Sakurako; Shibata, Nozomu; Kakeshita, Hiroshi; Kodama, Hiroshi; Igarashi, Kazuaki; Takimura, Yasushi

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Keywords Hits: cellulase: 144, reesei: 25, cbh1: 10, xyr1: 136

Keywords Frequency (TF): cellulase: 0.019; reesei: 0.003; cbh1: 0.001; xyr1: 0.018

Abstract:

Trichoderma, bc toggle=yesreesei is a widely used host for producing cellulase and hemicellulase cocktails for lignocellulosic biomass degradation. Here, we report a genetic modification strategy for industrial T., bc toggle=yesreesei that enables enzyme production using simple glucose without inducers, such as cellulose, lactose and sophorose. Previously, the mutated XYR1V821F or XYR1A824V was known to induce xylanase and cellulase using only glucose as a carbon source, but its enzyme composition was biased toward xylanases, and its performance was insufficient to degrade lignocellulose efficiently. Therefore, we examined combinations of mutated XYR1V821F and constitutively expressed CRT1, BGLR, VIB1, ACE2, or ACE3, known as cellulase regulators and essential factors for cellulase expression to the T., bc toggle=yesreesei E1AB1 strain that has been highly mutagenized for improving enzyme productivity and expressing a -glucosidase for high enzyme performance. The results showed that expression of ACE3 to the mutated XYR1V821F expressing strain promoted cellulase expression. Furthermore, co-expression of these two transcription factors also resulted in increased productivity, with enzyme productivity 1.5-fold higher than with the conventional single expression of mutated XYR1V821F. Additionally, that productivity was 5.5-fold higher compared to productivity with an enhanced single expression of ACE3. Moreover, although the DNA-binding domain of ACE3 had been considered essential for inducer-free cellulase production, we found that ACE3 with a partially truncated DNA-binding domain was more effective in cellulase production when co-expressed with a mutated XYR1V821F. This study demonstrates that co-expression of the two transcription factors, the mutated XYR1V821F or XYR1A824V and ACE3, resulted in optimized enzyme composition and increased productivity.

Summary:

Introduction: Lignocellulosic biomass is the most abundant sustainable feedstock for biorefinery and biofuel production. Therefore, effectively utilizing lignocellulosesic biomass could assist in addressing the problem. Extracellular cellulases and hemicellulase for lignochellulusic biomass degradation are produced mainly by filamentous fungi. Especially the filamentous filamentous yeast Trichodermareesei(teleomorph Hypocreajecorna) is a well-known microorganism that produces large quantities of mixed extracellular cellulase and hemicellulase enzymes for the degradation of lignoscellulusc biomass.

Results: The constitutive expression of mutated XYR1V821Fand its candidate cooperative factors was combined with the disruption of repressors. The disruption of theace1andrce1repressorestrains was found to reduce protein productivity. The high-intensity band indicated by an open arrow was found in Fig.b. This protein was found as a glycoside hydrolase family 55 (GH55) protein (Trire2121746) by nano LCMS/MS analysis. This indicates that little cellulase production was observed in the three strains with glucose as the sole carbon source.

Discussion: Enhanced cellulase productivity using glucose as the sole carbon source was achieved by constitutive expression of mutated XYR1 V821F or XyR1 A824V and ACE3, especially ACE3 having a partially truncated DNA-binding domain. Previously, the deletion of these factors was reported to cause loss or reduction of celllulase expression,. These factors were also affected by enhanced expression of T. reesei.

Figures:

Figure 1: Effects of constitutive expression of V821F mutated XYR1 and disruption of repressors on glucose culture. (a) Extracellular protein from the cultivation of samples after 72, 96, and 120h. Trichoderma reesei strain E1AB1 was cultivated in shake flasks on an inducing medium containing 3% cellulose, and recombinant strains E1AB1 ace1 , E1AB1 rce1 , E1AB1-X (ace1 -P act1 - xyr1 V821F), and E1AB1-X rce1 were cultivated on a non-inducing medium containing 3% glucose. (b) SDS-PAGE analysis of the secreted proteins after 72h. T. reesei strain E1AB1 with 3% cellulose (lane 1) and 3% glucose (lane 2) cultures, and recombinant strains E1AB1 ace1 (lane 3), E1AB1 rce1 (lane 4), E1AB1-X (lane 5), and E1AB1-X rce1 (lane 6) with 3% glucose cultures. Open arrowhead indicates the confirmed in glucose culture (lanes 24). Closed arrowheads indicate clearly overexpressed proteins, likely corresponding to the main xylanase (BXL1, XYN1, XYN2, lanes 5 and 6). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S4. Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. p 0.01.

Figure 2: Co-expression of factors involved in cellulase transcription in the E1AB1-X strain. Trichoderma reesei strain E1AB1-X was co-expressed with various factors that relate to cellulase expression and cultivated in shake flasks on a non-inducing medium containing 3% glucose. (a) Extracellular protein from cultivation samples after 96h. (b) SDS-PAGE analysis of the secreted proteins after 72h in a glucose culture of T. reesei E1AB1-X (lane

1, ace1 -P act1 - xyr1 V821F , as control), E1AB1-XC (lane 2, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - crt1), E1AB1-XB (lane 3, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - bglr), E1AB1-XV (lane 4, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - vib1), E1AB1-XA2 (lane 5, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace2), and E1AB1-XA3 (lane 6, E1AB1-XA3, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (PT)). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S5. (c) Volumetric enzyme activity of the 72-h culture supernatant. One unit of activity is defined as the amount of enzyme that produced 1mol of p -nitrophenol per minute per mL of culture supernatant from the substrate at 50C. Bar graphs show the activity relative to that of E1AB1-X. Error bars indicate standard deviations. Statistical significance was determined by the two-tailed unpaired Students t -test. p 0.01.

Figure 3: Confirmation of the genetic combinations required for high cellulase production under inducer-free conditions. (a) Putative domain of XYR1 and ACE3 used in the study, amino acid mutations in XYR1, and the presumed variant of ACE3. Shaking flask cultivations were performed using 3% cellulose: Lane 1 and 3% glucose: Lane 29. The strains used in this figure were Lane 1 and 2: E1AB1, Lane 3: E1AB1-X (ace1 -P act1 - xyr1 V821F), Lane 4: E1AB1-A3 (rce1 -P act1 - ace3 (PT)), Lane 5: E1AB1-XA3 (ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (PT)), Lane 6: E1AB1-XA3fl (ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (FL)), Lane 7: E1AB1-XA3nhr (P act1 - xyr1 V821F and P act1 - ace3 (PT) by non-homologous recombination), Lane 8: E1AB1-X824A3nhr (P act1 - xyr1 A824V and P act1 - ace3 (PT) by non-homologous recombination) and Lane 9: E1AB1-XwtA3nhr (P act1 - xyr1 WT and P act1 - ace3 (PT) by non-homologous recombination).The carbon sources, strains, and genotypes used in this figure were organized in an Additional file, Table S3. (b) SDS-PAGE of secreted proteins after 72h cultivation (2.5g-protein/lane). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S6. (c) Extracellular protein concentration at 96h. Relative transcription levels of xyr1 (d) and ace3 (e). The real-time PCR was performed on the samples taken after 48h. The transcriptional levels of pgk1 and E1AB1 strain on glucose culture were measured for reference calculation and data normalization, respectively, and analyzed using the Ct method. Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. a indicated p0.01 for the test with Lane 5 (E1AB1-XA3) and b indicated p0.01 for the test with Lane 7 (E1AB1-XA3nhr).

Figure 4: Analysis of gene expression, enzyme activity, composition, and saccharification of microcrystalline cellulose. (a) Relative gene expression of the major cellulases and xylanases. The real-time PCR was performed on the 48h culture sample with glucose as a carbon source. The transcriptional levels of pgk1 and E1AB1-X strain RNA were measured for reference calculation and data normalization, respectively, and analyzed using the Ct method. (b) The volumetric activity of each enzyme in the culture supernatant obtained after 72h of cultivation with glucose as the carbon source. One unit of activity is defined as the amount of enzyme that produced 1mol of p-nitrophenol per minute per mL culture of the supernatant from the substrate at 50C. Bar graphs show relative activity to E1AB1-X. (c) Concentration of each enzyme component was calculated from the enzyme composition and total secreted protein concentration. The compositions were calculated from the band patterns following SDS-PAGE (see Additional file Fig. S2) derived from E1AB1 (culturing using cellulose: C and glucose: G), E1AB1-X (G), E1AB1-A3 (G), and E1AB1-XA3 (G) strains. Protein concentrations were determined from the 96-h value. (d) Microcrystalline cellulose saccharification using the same protein dosage (2.0mg protein/g cellulose). (e) Microcrystalline cellulose saccharification using the same volume of culture supernatant (0.87mL culture supernatant/g cellulose). Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. p 0.01, p 0.05.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, ARE1, BGLR, BXL1, CBH, CBH1, CBH2, CIP2, CRE1, CRT1, CRZ1, CTF1, Cellulase, EG, EG1, Gal4, LAE1, MP, PAC1, RCE1, STR1, VEL1, VIB1, WT, XYN1, XYN2, XYR1, ace1, ace2, act1, cbh1, cbh2, cellulase, cre1, crt1, egl1, pgk1, pyr4, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: tube

Pathways: N/A

Query Result 16/25

Deciphering the Cis-Regulatory Elements for XYR1 and CRE1 Regulators in Trichoderma reesei, bIdentification of XYR1 and CRE1 Binding Sites in Trichoderma reesei

Silva-Rocha, Rafael; Castro, Lilian dos Santos; Antonito, Amanda Cristina Campos; Guazzaroni, Mara-Eugenia; Persinoti, Gabriela Felix; Silva, Roberto Nascimento; Bader, Joel S.

PLoS ONE

Year: 2014 PMID: 24941042 Citations: 21 Score: 96.49 DOI: 10.1371/journal.pone.0099366

Keywords Hits: cellulase: 37, reesei: 26, cbh1: 0, xyr1: 50

Keywords Frequency (TF): cellulase: 0.011; reesei: 0.008; cbh1: 0.000; xyr1: 0.015

Abstract:

In this work, we report the in silico identification of the cis-regulatory elements for XYR1 and CRE1 proteins in the filamentous fungus Trichoderma reesei, two regulators that play a central role in the expression of cellulase genes. Using four datasets of condition-dependent genes from RNA-seq and RT-qPCR experiments, we performed unsupervised motif discovery and found two short motifs resembling the proposed binding consensus for XYR1 and CRE1. Using these motifs, we analysed the presence and arrangement of putative cis-regulatory elements recognized by both regulators and found that shortly spaced sites were more associated with XYR1- and CRE1-dependent promoters than single, high-score sites. Furthermore, the approach used here allowed the identification of the previously reported XYR1-binding sites from cel7a and xyn1 promoters, and we also mapped the potential target sequence for this regulator at the cel6a promoter that has been suggested but not identified previously. Additionally, seven other promoters (for cel7b, cel61a, cel61b, cel3c, cel3d, xyn3 and swo genes) presented a putative XYR1-binding site, and strong sites for CRE1 were found at the xyr1 and cel7b promoters. Using the cis-regulatory architectures nearly defined for XYR1 and CRE1, we performed genome-wide identification of potential targets for direct regulation by both proteins and important differences on their functional regulons were elucidated. Finally, we performed binding site mapping on the promoters of differentially expressed genes found in T. reesei mutant strains lacking xyr1 or cre1 and found that indirect regulation plays a key role on their signalling pathways. Taken together, the data provided here sheds new light on the mechanisms for signal integration mediated by XYR1 and CRE1 at cellulase promoters.

Summary:

Introduction: Trichoderma reeseiis a filamentous fungus extremely relevant to biotechnology due to its remarkable capability to produce a wide number of cellulolytic enzymes. This mesophilic organism is endowed with a tremendous repertoire of hydrolytic enzyme related to the deconstruction of lignocellulosic biomass from plants that are of high importance for biotech processes such as paper industry or fuel production. Due to its elevated biotechnological potential, T. reseis been extensively studied in the past decades as a model of celluases and hemicellulases producing organism.

Results and Discussion: XYR1 and CRE1 inT. reesei are characterized by short repetitive DNA motifs. We used four different datasets of co-regulated genes to search for short repetitive motifs potentially recognized by these regulators. We found that motifs resembling the reported consensus for XyR1 (5-GGCWWW-3, where W stands for A or T) and Cre1 (5 -SYGGRG-3,where S stands for C or T and R for A and G) were found in all four datasets.

Conclusions: XYR1 and CRE1 proteins are a pair of general regulators that coordinate the expression of cellulase-encoding genes in T. reesei. The data provided here addressed for the first time the quantitative identification of binding sites for Xyr1 and XRE1 protein, two general regulator that coordinates the expression ... ... The current proposed induction mechanism suggests that under a repression condition (i.e., in the presence of glucose) the production of cellulases is completely blocked, while under starvation conditions basal levels of these enzymes are produced. Subsequently, when the fungus finds cellulose, the produced enzymes act on this insoluble substrate to generate soluble inducers such as sophorose, which in turn would trigger the signal for high level of cellular production.

Figures:

Figure 1: Schematic representation of the approach used for motif discovery in T. reesei .

Figure 2: Motifs identified in the cellulose and glucose regulated genes.

Figure 3: Search for single XYR1 and CRE1 binding sites on different promoter datasets.

Figure 4: Search for repeated XYR1 and CRE1 binding sites on different datasets.

Figure 5: Identification of XYR1 and CRE1-binding sites at target promoters.

Figure 6: Defining the categories of genes potentially regulated directly by XYR1 and CRE1.

Mentioned biomedical entities:

Genes: ACE1, ACE2, BXL1, BglR, C, CRE1, Cel12a, Cel1a, Cel5a, Cel61a, Cel7a, Cel7b, Cellulase, FMRP, Mig1, PWM, XYN1, XYN2, XYR1, XYR1 , cbh2, cel3c, cel6a, cellulase, cre1, xyn1, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: T. reesei

Tissues: N/A

Pathways: N/A

Query Result 17/25

The VELVET A Orthologue VEL1 of Trichoderma reesei Regulates Fungal Development and Is Essential for Cellulase Gene Expression

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PLoS ONE

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Keywords Frequency (TF): cellulase: 0.011; reesei: 0.013; cbh1: 0.000; xyr1: 0.003

Abstract:

, bTrichoderma reesei is the industrial producer of cellulases and hemicellulases for biorefinery processes. Their expression is obligatorily dependent on the function of the protein methyltransferase LAE1. The Aspergillus nidulans orthologue of LAE1 - LaeA - is part of the VELVET protein complex consisting of LaeA, VeA and VelB that regulates secondary metabolism and sexual as well as asexual reproduction. Here we have therefore investigated the function of VEL1, the T. reesei orthologue of A. nidulans VeA. Deletion of the T. reesei vel1 locus causes a complete and light-independent loss of conidiation, and impairs formation of perithecia. Deletion of vel1 also alters hyphal morphology towards hyperbranching and formation of thicker filaments, and with consequently reduced growth rates. Growth on lactose as a sole carbon source, however, is even more strongly reduced and growth on cellulose as a sole carbon source eliminated. Consistent with these findings, deletion of vel1 completely impaired the expression of cellulases, xylanases and the cellulase regulator XYR1 on lactose as a cellulase inducing carbon source, but also in resting mycelia with sophorose as inducer. Our data show that in T. reesei VEL1 controls sexual and asexual development, and this effect is independent of light. VEL1 is also essential for cellulase gene expression, which is consistent with the assumption that their regulation by LAE1 occurs by the VELVET complex.

Summary:

Introduction: The lignocellulose degradation enzymes in the plant cellulolytic bacterium Trichoderma reesei are regulated by the protein methyltransferase LAE1, which is a homologue of the protein LaeA. This regulation is mediated by a methylation of H3K4 and H3k9 in the chromatin. This methylated chromatine is required for the regulation of the enzymes.

Results: The VEL1 orthologue ofT. reesei contains a single copy of thevel1gene. The ORF ofvel1is interrupted by a 79 bp intron and encodes a 574 amino acid protein. The sequence ofvel1, which is similar to the VeA orthologues inT.virensandT. atroviride, was phylogenetically characterized. The protein is capable of entering the nucleus and contains sulfonylated arginine and glutamine.

Discussion: In this paper, we have explored the function of the velvet gene vel1 of T. reesei. While this global regulator is conserved in Pezizomycota, studies of veA orthologs across several fungal genera have now established a significant diversity in its impact on fungal development , The most striking example is asexual sporulation, a trait influenced by VeA/VelA/VEL1 in all fungi: whereas aveA/velAknock out in A. nidulans, P. chrysogenum and N. crassaincreases conidiation it results in decreased conidiaton in the corresponding knock-out mutants of A. fumigatus, A. parasiticus,A. flavus, Fusarium fujikuroi, F. graminearum, Dothistroma septosporum and T. virens . Our study shows that T.

Figures:

Figure 1: Effect of carbon source and light on vel1 transcript levels in T. reesei .

Figure 2: The impact of VEL1 on morphology of T. reesei in submerged culture.

Figure 3: VEL1 is required for T. reesei development.

Figure 4: The impact of VEL1 on the utilization of different carbon sources by T. reesei .

Figure 5: Impact of vel1 deletion or overexpression on growth and cellulase formation by T. reesei on lactose.

Figure 6: Impact of vel1 deletion or overexpression on transcript levels of two cellulase, one hemicellulase and a cellulase regulator gene in T. reesei during growth on lactose.

Figure 7: Impact of vel1 deletion or overexpression on transcript levels of two cellulase, one hemicellulase and a cellulase regulator gene in T. reesei upon induction by sophorose.

Mentioned biomedical entities:

Genes: LAE1, LaeA, MP, REST, RUT, VEL1, VeA, VelB, XYR1, amylase, cel6a, cellulase, ctr1, lae1, mat1-1, mat1-2, tef1, vel1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. flavus, A. fumigatus, A. nidulans, Escherichia coli, Human, T. reesei, T. virens, humans

Tissues: N/A

Pathways: N/A

Query Result 18/25

Tailoring the expression of Xyr1 leads to efficient production of lignocellulolytic enzymes in Trichoderma reesei for improved saccharification of corncob residues

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Biotechnology for Biofuels and Bioproducts

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Keywords Hits: cellulase: 96, reesei: 50, cbh1: 28, xyr1: 124

Keywords Frequency (TF): cellulase: 0.015; reesei: 0.008; cbh1: 0.004; xyr1: 0.019

Abstract:

The filamentous fungus Trichoderma reesei is extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulase gene expression. However, the role of Xyr1 in the regulation of cellulase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolytic enzyme production.

Summary:

Background: The filamentous fungus Trichoderma reeseiis extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulolytic enzyme gene expression. However, the role of Xr1 in the regulation of cellululase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolysis enzyme production.

Results: The xyr1 gene was overexpressed in T. reesei by cellulose-inducible promoters Pegl2, Pcbh1, and Pcdna1 promoter. The maximum transcript levels ofxyre1were found in the QCDX strain, followed by QCBX and QE2X. Strains were cultured at 30 and 200rpm in flasks using the CPM medium with 2 (w).

Conclusions: Our results underpin that the precise tailoring expression of xyr1is essential for highly efficient cellulase synthesis, which provide new insights into the role of Xyr 1 in regulating cellulolytic enzyme expression in T. reesei. Moreover, these results also provides a prospective strategy for strain improvement to enhance the lignocellulolysis enzyme production for use in biorefinery applications. 2016 Elsevier B.V.

Introduction: Lignocellulosic biomass is an abundantly available, inexpensive organic resource on the earth, which is basically composed of cellulose, hemicellulose, and lignin . Biological degradation of lignocellulocellulose biomass by lignochelolytic enzymes into fermentable sugars for the production of biofuels and other biochemicals has received extensive attention ,. However, the high cost of cellulase production, which was the major barrier for practical applications, limits the development of an efficient and economically viable lignoscellulusic feedstock based biorefinery . The filamentous fungus Trichoderma reeseiencodes a wide repertoire of cellobiohydrolases (CBHs), endoglucanases (EGs), and -glucosidase (exactly BGL1, EC 3.2.1.21) to degrade cellulose and other biomass components, which makes it be extensively employed in the industrial cellulolytic production .

Discussion: In T. reesei, genetic engineering of transcriptional regulators has been used as a powerful tool for enhancing cellulase production. Xyr1 is the key transcriptional activator of lignocellulolytic genes, whose overexpression can improve cellullase yields under both inducing and non-inducing conditions. However, the effect of xyr1/dosage regulation on cellulolase expression has not yet been elucidated. Herein, the expression of XYR1was tailored under promoters of different strengths and we found that the xYR1/overexpression with the Pegl2promoter resulted in a substantial improvement in cellululases production and saccharification efficiency. In previous studies, the promoter Ppdcwas chosen to drivexyR1expression for the construction of cellulesa hyperproducing strain ,,. In addition, the Promoter Ptcu1was also used to the expression , and the resultant strain could achieve full cellullasa expression . By roughly comparing the effects of the two promoter, we speculate that the overexpression ofxy1driven by the stronger promoter may have a more pronounced influence on cellulase production. However this does not imply that the maximization of cellulose production.

Conclusion: In the present study, we showed that the Pegl2-driven xyr1overexpression significantly improved production of lignocellulolytic enzymes in T. reesei. The total cellulase activity of QE2X was up to 8.4IU/mL under inducing condition. The glucose release from ACR by the enzyme mixture of Q E2X were improved by 57.2. To the best of our knowledge, this is the first report regarding the roles ofxyregulatory level on cellulolysis in T . eesii.

Figures:

Figure 1: Construction and validation of the xyr1 overexpression strains. A Schematic diagrams of the expression cassettes used for differential overexpression of the xyr1 gene. The overexpression of xyr1 was controlled by three different promoters (P egl2 , P cbh1, and P cdna1), respectively. B Transcript levels of xyr1 which was determined by RT-qPCR in QEB4 and three independent transformants of each xyr1 overexpression strain (QE2X, QCBX, and QCDX). Strains were cultivated at 30 and 200rpm in flasks using the CPM medium with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript level was normalized to that of actin (t -test, p 0.01). Results are means of three biological replicates and error bars indicateSD

Figure 2: Detection of the cellulase secretion by T. reesei QEB4 and the xyr1 overexpression strains in different culture plates. The strains were assessed on the MCC plate (A), the CMC plate (C), and the CMC-esculin plate (E) to evaluate the secretion capacity of total cellulase, EGs and BGL, respectively. B The ratios between the diameter of the transparent zones and the colonies on MCC plate. D The hydrolytic halo diameters of the strains on CMC-Na plate. F The black zone diameters of the strains on CMC-esculin plate. p 0.05; p 0.01 (t -test). Results are means of three biological replicates and error bars indicateSD

Figure 3: Transcript analysis by RT-qPCR for genes encoding cellulases and regulators. Genes encoding cellulases (A) and transcription factors (B) were analyzed in the xyr1 overexpression strains with T. reesei QEB4 as the control. Strains were cultivated at 30 and 200rpm in flasks using the CPM medium with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 4: Transcript and CHART analysis of T. reesei QEB4 and the xyr1 overexpression strains. The genes of cbh1 (A), cbh2 (B), and the native xyr1 (C) were determined by RT-qPCR and CHART-PCR, respectively. The strains were cultured in CPM with 2% (w/v) Avicel for 48h. All data were normalized to the reference gene actin. The transcript levels are depicted on the x-axis and the chromatin accessibility indices (CAI) are plotted on the y-axis. Results are means of three biological replicates and error bars indicateSD

Figure 5: Transcript analysis of the genes encoding the ER-associated components by RT-qPCR. The UPR- (A) and ERAD- (B) related genes were analyzed in the xyr1 overexpression strains with T. reesei QEB4 as the control. Strains were cultivated at 30 and 200rpm in flasks using CPM with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 6: Cellulase activities and total extracellular proteins of T. reesei QEB4 and the xyr1 overexpression strains. A FPase activity (FPA), B cellobiohydrolase (CBH) activity, C endoglucanase (EG) activity, D -glucosidase (BGL) activity, and E the total extracellular protein of the fermentation supernatant from QEB4 and three recombinant strains cultured on 2% (w/v) Avicel for 7days. p 0.05; p 0.01, ns=not significant (t -test). Results are means of three biological replicates and error bars indicateSD

Figure 7: Saccharification of different pretreated corncob residues by T. reesei QEB4 and QE2X. Saccharification of acid-pretreated corncob residue (A) and delignified corncob residue (B) with the same FPA dosage. p 0.01, ns=not significant (t -test). Results are means of 3 biological replicates and error bars indicateSD

Figure 8: Transcript analysis for genes encoding accessory proteins of T. reesei QEB4 and QE2X by RT-qPCR. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 9: Schematic summary showing the proposed model for the dosage regulation of xyr1 in T. reesei . A Compared to the strong xyr1 overexpression with the cbh1 / cdna1 promoter, the moderate xyr1 overexpression with the egl2 promoter can result in more open chromatin and stronger expression of cellulase gene, thus significantly enhancing cellulase production. B The lignocellulolytic enzyme production and saccharification efficiency were significantly improved in T. reesei QE2X, because of the high expression of cellulase and accessory proteins

Mentioned biomedical entities:
Genes: Ace1, Ace2, Ace3, B, BGL1, BglR, CBH, Cip1, Cip2, Cre1, Crz1, Ctf1, Dc, Ds, EG2, ER, FPA, Pac1, Ppdc, Rce1, Sec16, Swo1, Vib1, Xyr1, ace1, ace2, actin, bgl1, cbh1, cbh2, cellulase, cip1, cip2, cre1, ctf1, egl1, histone, hrd1, pdi1, ptrA, pyr4, swo1, vib1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 19/25

From induction to secretion: a complicated route for cellulase production in Trichoderma reesei

Yan, Su; Xu, Yan; Yu, Xiao-Wei

Bioresources and Bioprocessing

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Keywords Frequency (TF): cellulase: 0.028; reesei: 0.011; cbh1: 0.000; xyr1: 0.004

Abstract:

The filamentous fungus Trichoderma reesei has been widely used for cellulase production that has extensive applications in green and sustainable development. Increasing costs and depletion of fossil fuels provoke the demand for hyper-cellulase production in this cellulolytic fungus. To better manipulate T. reesei for enhanced cellulase production and to lower the cost for large-scale fermentation, it is wise to have a comprehensive understanding of the crucial factors and complicated biological network of cellulase production that could provide new perspectives for further exploration and modification. In this review, we summarize recent progress and give an overview of the cellular process of cellulase production in T. reesei, including the carbon source-dependent cellulase induction, complicated transcriptional regulation network, and efficient protein assembly and trafficking. Among that, the key factors involved in cellulase production were emphasized, shedding light on potential perspectives for further engineering.

Summary:

Introduction: The cellulolytic fungus Trichoderma reeseiis one of the most widely applied microorganisms for cellulase production. Approximate 100g/L extracellular cellulse could be produced inT. rese. The secreted celllulase mainly consists of two major cellobiohydrolases CBHI/CEL7A and CBBI/CELL6A, endoglucanases EGI/CELI7B and EGII/CELLE5A, and -glucosidases BGLI/CELE3A which account for 90 extracellular protein (de Paula et al). Besides, the hemicellulases, LPMOs (lytic polysaccharide monooxygenases), CIP (cellulose-induced protein) and swollenin which belong to the secretome ofT.

Figures:

Figure 1: Overview of cellulase induction in T. reesei . This figure shows the process of cellulase induction on cellulose and lactose. Initially, the cellulase CEL7A/6A, CEL7B/5A and -glucosidase CEL3A are involved in the degradation of cellulose, and BGA is essential for the cleavage of lactose. Next, the released mono- and disaccharides could be transported intracellularly through sugar transporters and participate in different metabolic pathways. Among that, different -glucosidases are critical for efficient cellulase production either on cellulose or lactose, through mediating the accumulation and formation of inducer. The key components and factors are depicted in corresponding areas

Figure 2: Cellulase regulation at the transcription level. In T. reesei , transcription of cellulase is under complicated and coordinated regulation networks including transcription factors, signal transduction, chromatin status, secondary metabolism and environmental factors such as light, nitrogen source, pH and metal ion, which are depicted in different areas with diverse colors. Only the factors involved in the crosstalk with other factors and pathways are depicted in the picture. The inter-regulation of different factors and regulators is shown by black arrows and lines. The dashed lines indicate characterized regulation mechanism in other fungi but not in T. reesei

Figure 3: Secretion pathway in T. reesei . Secreted cellulase will enter the classical secretion pathway, which passes through the ribosome, endoplasmic reticulum, Golgi, and secrets extracellularly. The ER lumen is amplified to give a comprehensive review of the protein assembly and ER quality control. The basic pathway for cellulase secretin is annotated as red arrow, with the assistance of multiple factors, including the PDI1, BiP1, HAC1, SNC1, etc. Insufficient folding or degradation of the secreted protein is indicated by gray arrows. Moreover, the promising secretion factors for improved cellulase production in T. reesei are also annotated aside, which need further exploration. The cellular structures are marked by purple, and the promising secretion pathways for engineering are shown with red font

Mentioned biomedical entities:

Genes: ACE2, ACE3, ALP, ARE1, BGA, BiP1, CEL1A, CEL1B, CEL3A, CEL6A, CEL7A, CLP1, CLR1, CLR2, CRE1, CRT1, CRZ1, CTF1, Cellulase, CrzA, ER, GNA-3, HAC1, IRE1, Kar2, LAE1, LaeA, MAPK, PAC1, PDI1, PKA, PMR1, PacC, RCE1, Rho, SAR1, SEC16, SLY1, SNC1, SNC2, SSO1, STP1, STR1, STR3, Su, VEL1, VEL2, VIB1, VeA, VelB, VelC, XYR1, YPR1, bgl1, cel1b, cellulase, cre1, cwp2, hac1, p24, pdi1, rac1, sar1, stp1, swo1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. nidulans, T. reesei

Tissues: bud

Pathways: N/A

Query Result 20/25

Development of a powerful synthetic hybrid promoter to improve the cellulase system of Trichoderma reesei for efficient saccharification of corncob residues

Wang, Yifan; Liu, Ruiyan; Liu, Hong; Li, Xihai; Shen, Linjing; Zhang, Weican; Song, Xin; Liu, Weifeng; Liu, Xiangmei; Zhong, Yaohua

Microbial Cell Factories

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Keywords Frequency (TF): cellulase: 0.006; reesei: 0.008; cbh1: 0.018; xyr1: 0.001

Abstract:

The filamentous fungus Trichoderma reesei is a widely used workhorse for cellulase production in industry due to its prominent secretion capacity of extracellular cellulolytic enzymes. However, some key components are not always sufficient in this cellulase cocktail, making the conversion of cellulose-based biomass costly on the industrial scale. Development of strong and efficient promoters would enable cellulase cocktail to be optimized for bioconversion of biomass.

Summary:

Background: A synthetic hybrid promoter was constructed and applied to optimize the cellulolytic system of Trichoderma reeseifor efficient saccharification towards corncob residues. Firstly, a series of 5 truncated promoters in different lengths were established based on the strong constitutive promoter Pcdna1. The strongest promoter amongst them was Pccd1(640 to 1bp upstream of the translation initiation codon ATG), exhibiting a 1.4-fold higher activity than that of the nativecda1promoter. Meanwhile, the activation region (821 to 622bpupstream ofthe translation insertion codon and devoid of the Cre1-binding sites) of the strong inducible promoterPcbh1was cloned and identified to be an amplifier in initiating gene expression. Finally, this activation regions was fused to the strongest promoteer Pcb1d, generating the novel synthetic hybrid Promoter P.cc. This engineered promoter DROVE strong gene expression by displaying 1.6-fold.

Results: TEXT: Characterization of a series of 5 truncated promoters ofcdna1As a strong constitutive promoter inT. reesei, Pcdnam1has been successfully utilized to express cellulases using glucose as the carbon source . Here, the 928-bp promoter region upstream of ATG of the genecdm1was analyzed as shown in Fig.. The promoter Pcm1contains several corecis-acting elements, such as TATA box, CCAAT box and GC box, which are essential elements to initiate transcription or enhance the transcriptional level . To functionally validate thecdam1promoter, five 5 thuncated promors in different lengths were established based on Pcna1. Five 5 htuncted promoter, Ppcda1-1(928bp), Ppdnam1-2(794bp) Pppcnam1-3(640bp, 490bp and 247bp).

Conclusions: In this work, a synthetic hybrid promoter Pccwas created and confirmed to direct high-level expression of the essential cellulase components BGLA and EG2 to achieve the optimized cellulolytic enzyme system in T. reesei. Furthermore, when applied to the saccharification of differently pretreated corncob residues, this cellulolar enzyme system exhibited the prominent performance that displayed 2.3-fold higher saccharinefficacy than that of the parental strain. The chimeric promoter constructed here was the first hybrid promoteer applied in T . eesiand raised the possibility to generate powerful tunable promoters.

Introduction: The filamentous fungus Trichoderma reeseihas been widely used as an industrial workhorse for cellulase production due to its extraordinary ability to produce large amounts of extracellular cellulolytic enzymes. Among the cellullase enzymes secreted by this fungi, the cellobiohydrolase 1 (CBH1) accounts for more than 60 of total extracellular protein , thus thecbh1promoter (PcbH1) was generally used as a strong inducible promoter for gene expression .

Discussion: This study created a novel hybrid promoter Pccusing the synthetic approach in T. reesei, which enabled higher gene expression than that of the powerful promoters Pcbh1and Pcdna1. In T.reeei cost-effective cellulolytic enzyme system is needed to achieve efficient bioconversion of lignocellulosic biomass, and it could be obtained by modifying strains through improving cellulase expression levels and optimizing enzyme system ,. Promoter development is an essential method for strain modification and metabolic engineering . Thus, this study createda novel promoterPcc using the synthetic .

Figures:

Figure 1: Nucleotide sequences of the P cdna1 (a) and P cbh1 (b). The nucleotides ahead of A of translation initiation site (ATG) of cdna1 and cbh1 are numbered as 1. Known cis -acting elements in P cdna1 and mainly in the 869bp to 621bp region of P cbh1 are shown in different color. The descriptions of elements (TATA box, CAAT box, GC box, Xyr1-binding site, Ace2-binding site and Cre1-binding site) are shown in Additional file 1 : Table S1. An arrow above the sequence indicates the start point of different truncated fragments

Figure 2: Identification of the truncated fragments of P cdna1 fused with cbh1 as the reporter gene. a Strategy of targeted replacement of P cbh1 with the truncated fragments of P cdna1 fused with cbh1 . b , c CBH activity of the transformants PD1-1 (P cdna1-1), PD1-2 (P cdna1-2), PD1-3 (P

cdna1-3), PD1-4 (P cdna1-4), PD1-5 (P cdna1-5) and the parental strain QM53 (control) cultured in GMM for 5days (b) and AMM for 7days (c). Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 3: Identification of P cbh1 activation region (AR) using egfp as the reporter gene. a Strategies of the construction of P cbh1-dc (P cbh1 devoid of Cre1-binding sites) and P cbh1-cr (core promoter of P cbh1) fused with egfp . RR: regulation region; CR: core region; AR: activation region. b , c Fluorescence relative ratio between the transformants PBD1 (P cbh1-dc-egfp), PBC1 (P cbh1-cr-egfp) and the parental strain QP4 cultured in GMM for 2days (b) and AMM for 3days (c). Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 4: Construction of hybrid promoter P cc . a Strategies of the construction of P cc fused with the reporter gene egfp . b Fluorescence relative ratio between the transformants PB1 (P cbh1-egfp), PD1 (P cdna1-egfp), PCC and the parental strain QP4, respectively. Strains were cultured in AMM for 2days. c The fluorescence of transformant PCC (P cc-egfp) and the parental strain QP4 cultured with glucose for 24h and Avicel for 3days were observed under the fluorescence microscope. The results shown represent one of at least three independent experiments. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 5: Expression of bglA driven by P cc resulted in increase of BGL activity and FPA. Transformants expressing bglA driven by P cc and the parental strain QP4 cultured on glucose-esculin plates and CMC-esculin plates at 30 for 2days (a). SDS-PAGE analysis of the total extracellular cellulases of transformants and QP4 were shown after cultured in GMM for 5days (b) and CPM for 7days (c). BGLA bands were indicated by arrows. BGL activity (d) and FPA (e) of the transformants and QP4 were measured after cultured in GMM for 5days and CPM for 7days. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 6: Expression of eg2 driven by P cc resulted in increase of EG activity and FPA. SDS-PAGE analysis of the total extracellular cellulases of transformants and the parental strain QP4 were shown after cultured in GMM for 3days (a) and CPM for 7days (b). EG2 bands were indicated by arrows. EG activity (c) and FPA (d) of the transformants and QP4 were measured after cultured in GMM for 5days and CPM for 7days. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 7: Co-overexpression of bglA and eg2 driven by P cc in T. reesei resulted in increase of FPA. The transformants and the parental strain QP4 were cultured in CPM for 7days. FPA were measured at the end of cultivation. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 8: Saccharification of different pretreated corncob residues using the extracellular enzymes from QP4 and the transformants co-overexpressing bglA and eg2 driven by P cc . a Glucose released from acid-pretreated corncob residue at 48h. b Glucose released from delignified corncob residue at 48h. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Mentioned biomedical entities:

Genes: AR, Ace1, Ace2, BGL1, BGLA, CBH, CBH1, Cre1, EG2, FPA, Hap5, PBC1, PBD1, PD1, Xyr1, bgl1, cbh1, cellulase, gpdA, hph, pdc, ptrA, pyrG

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 21/25

Systems biological approaches towards understanding cellulase production by Trichoderma reesei

Kubicek, Christian P.

Journal of Biotechnology

Year: 2013 PMID: 22750088 Citations: 50 Score: 78.54 DOI: 10.1016/j.jbiotec.2012.05.020

Keywords Hits: cellulase: 78, reesei: 68, cbh1: 1, xyr1: 3

Keywords Frequency (TF): cellulase: 0.017; reesei: 0.015; cbh1: 0.000; xyr1: 0.001

Abstract:
First systems biology review on Trichoderma reesei. First omics comparison of Trichoderma reesei and other Trichoderma spp. The omics results offer novel understanding of biomass hydrolyzing enzymes. Routes for further research are illustrated.

Summary:
Introduction: The plant cell wall consists of the -(1,4)-linked glucose polymer cellulose, hemicellulose polysaccharides of varying composition, and lignin. The former two are formed by plants at an annual production rate of about 7.2 and 6 10 10 tons, respectively. They thus represent the largest reservoir of renewable carbon sources on earth which could be used for the production of biofuels and other biorefinery products, thereby replacing products derived from fossile carbon components.

Concluding remarks: As I have attempted to highlight above, the application of several -omics to T. reeseiand its formation of biomass degrading enzymes have led to a number of new and exciting insights into this process and its regulation, and many more findings will be published in the next future. In addition, tools for high throughput molecular genetics work with T.reeeihave been considerably expanded over the last years, and recipient strains with almost 100 integration into the homologous locus, a high throughpput system for gene disruption, methods for marker removement, several dominant selection markers, and mating type partners for sexual crossing are now available (for review see). It is therefore possible to tackle all important questions related to cellulase expression and secretion directly in this organism, and consequently there is no need to perform these investigations in other model fungi. The only disadvantage, which remains and which probably will not be rapidly overcome, is the comparatively small community of researchers working on the molecule biology of T.

Figures:

Figure 1: CAZys, indicated by GH-number, that are amplified in Trichoderma. Numbers on the vertical axes of the plots indicate number of gene paralogues. Abbreviations : J, T. reesei ; V, T. virens ; T, T. atroviride ; A, Aspergillus nidulans ; N, Aspergillus niger ; G, Gibberella zeae ; M, Magnaporthe grisea ; P, Podospora anserina ; B, Botryodiplodia fuckeliana ; S, Sclerotium sclerotiorum ; E, Neurospora crassa ; O, Rhizopus nigricans ; D, Batrachochytrium dendrobatidis . Data were compiled from Martinez et al. (2008) , Eastwood et al. (2011) , Goodwin et al. (2011) , and Kubicek et al. (2011, 2012) .

Figure 2: Genomic synteny of areas containing cellulase clusters, as determined by REEF. Synteny data were retrieved from the synteny browser at http://genome.jgi-psf.org/Trire2/Trire2.home.html , using the genome of T. atroviride as a comparison. Interrupted bars indicate lack of synteny. The different degrees of fillings of the bars (from black to grey) indicate occurrence of the respective areas on different scaffolds in T. atroviride . Numbers indicate the respective position on the scaffold in Mbps. Areas containing cellulase genes are indicated by red boxes.

Figure 3: Pedigree of H. jecorina mutant strains whose genome has been sequenced. The parent strain, whose genome sequence is already available is boxed; the cellulase negative mutants are located within the blue box.

Figure 4: Extracellular CAZymes identified by proteomic approaches by Jun et al., 2011 (A), Herpol-Gimbert et al., 2009 (B) and Adav et al., 2011 (C). Black boxes indicate that the protein was detected, numbers are the protein IDs in the T. reesei genome database. Note that the endoglucanase marked with an is actually a cellulose monooxygenase.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ATF4, B, CBH1, CIP1, CPC1, CRE1, Cel6A, Cel7A, GH, GHs, H2A, H4, IRE1, LAE1, LaeA, MGST3, RIP, RUT, XYL1, XYR1, actin, cellulase, cpc1, cre1, gls2, histone, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: T. reesei, T. virens, mouse

Tissues: N/A

Pathways: N/A

Query Result 22/25

Interdependent recruitment of CYC8/TUP1 and the transcriptional activator XYR1 at target promoters is required for induced cellulase gene expression in Trichoderma reesei

Wang, Lei; Zhang, Weixin; Cao, Yanli; Zheng, Fanglin; Zhao, Guolei; Lv, Xinxing; Meng, Xiangfeng; Liu, Weifeng; Butler, Geraldine

PLoS Genetics

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Keywords Hits: cellulase: 74, reesei: 33, cbh1: 3, xyr1: 86

Keywords Frequency (TF): cellulase: 0.010; reesei: 0.005; cbh1: 0.000; xyr1: 0.012

Abstract:

Cellulase production in filamentous fungus Trichoderma reesei is highly responsive to various environmental cues involving multiple positive and negative regulators. XYR1 (Xylanase regulator 1) has been identified as the key transcriptional activator of cellulase gene expression in T. reesei. However, the precise mechanism by which XYR1 achieves transcriptional activation of cellulase genes is still not fully understood. Here, we identified the TrCYC8/TUP1 complex as a novel coactivator for XYR1 in T. reesei. CYC8/TUP1 is the first identified transcriptional corepressor complex mediating repression of diverse genes in Saccharomyces cerevisiae. Knockdown of Trcyc8 or Trtup1 resulted in markedly impaired cellulase gene expression in T. reesei. We found that TrCYC8/TUP1 was recruited to cellulase gene promoters upon cellulose induction and this recruitment is dependent on XYR1. We further observed that repressed Trtup1 or Trcyc8 expression caused a strong defect in XYR1 occupancy and loss of histone H4 at cellulase gene promoters. The defects in XYR1 binding and transcriptional activation of target genes in Trtup1 or Trcyc8 repressed cells could not be overcome by XYR1 overexpression. Our results reveal a novel coactivator function for TrCYC8/TUP1 at the level of activator binding, and suggest a mechanism in which interdependent recruitment of XYR1 and TrCYC8/TUP1 to cellulase gene promoters represents an important regulatory circuit in ensuring the induced cellulase gene expression. These findings thus contribute to unveiling the intricate regulatory mechanism underlying XYR1-mediated cellulase gene activation and also provide an important clue that will help further improve cellulase production by T. reesei.

Summary:

Introduction: CYC8/TUP1 corepressors are involved in the regulation of gene expression in eukaryotes. However, CY C8/ TUP1 has been implicated in gene activation in S.cerevisiae. Therefore, the role of CYc8/tup1 in gene regulation in S .cerrevisiaceae is not well understood. Therefore , the CYP8/TUP1 corepression complex is a key component of the eutrophic repression system.

Results: The CYC8/TUP1 complex is a stable complex in T.reesei. The phylogenetic analysis revealed that the CYc8/ TUP1 sequence orthologs were widely distributed among filamentous ascomycete fungi including a few well-known cellulolytic Trichoderma, Penicillium, Fusarium, Neurospora, and Aspergillus strains but none of these putative CY C8/C1 homologs have been assessed regarding their functional involvement in plant cell wall degradation.

Discussion: The results of this study show that a TrCYC8/TUP1 complex exists in T. reesei, and is required for high-level activation of cellulase genesin vivo. Similar to functions in other filamentous fungi, repression ofTrcyc8 or Trtup1results in gross defects in vegetative growth and asexual spore production in T rese. Unlike the carbon catabolite repressor CRE1, the TrCYc8/ TUP1 Complex is critical for the induced cellullase gene expression. ChIP analyses reveal that TrTUP1/XYR1 are interdependent for efficient binding at cellululases gene promoters to ensure their induced expression.

Figures:

Figure 1: Evolutionarily conserved CYC8/TUP1 complex exists in T . reesei .

Figure 2: Knockdown of Trcyc8 and Trtup1 by conditional repression.

Figure 3: Knockdown of Trcyc8 or Trtup1 compromised T . reesei growth and conidiation on agar plates but not in liquid media.

Figure 4: Knockdown of Trcyc8 and Trtup1 abolished cellulase production.

Figure 5: Trcyc8 or Trtup1 repression resulted in a significant decrease in the transcription of cellulase gene and xyr1 transcription.

Figure 6: TrTUP1 was recruited to cellulase gene promoters upon induction in an XYR1-dependent manner.

Figure 7: XYR1 overexpression failed to rescue the defective cellulase production in P tcu1 - Trcyc8 KD or P tcu1 - Trtup1 .

Figure 8: Efficient binding of XYR1 to cellulase genes in vivo relied on TrTUP1.

Figure 9: The effect of Trcyc8 or Trtup1 repression on the association of histone H4 with cellulase gene promoters upon cellulose induction.

Figure 10: A working model for interdependent binding of TrCYC8/TUP1 and XYR1 on cellulase gene promoters to mediate their induced expression.

Mentioned biomedical entities:

Genes: CBH1, CEL7A, CRE1, CYC8, FZ, GCN4, H4, IP, LexA, MA, MEL1, MIG1, MO, PIC, SD, SspI, TPR, TUP1, WD40, XYR1, Xyr1, actin, arg1, arg4, bgl1, cbh1, cellulase, cre1, cyc1, fre2, histone, pyr4, tup1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: T. reesei

Tissues: tissue

Pathways: N/A

Query Result 23/25

A Mitogen-Activated Protein Kinase Tmk3 Participates in High Osmolarity Resistance, Cell Wall Integrity Maintenance and Cellulase Production Regulation in Trichoderma reesei, bFunction of a Hog1-Type MAPK (Tmk3) in T. reesei

Wang, Mingyu; Zhao, Qiushuang; Yang, Jinghua; Jiang, Baojie; Wang, Fangzhong; Liu, Kuimei; Fang, Xu; Harris, Steven

PLoS ONE

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Keywords Frequency (TF): cellulase: 0.010; reesei: 0.020; cbh1: 0.001; xyr1: 0.000

Abstract:

The mitogen-activated protein kinase (MAPK) pathways are important signal transduction pathways conserved in essentially all eukaryotes, but havent been subjected to functional studies in the most important cellulase-producing filamentous fungus Trichoderma reesei. Previous reports suggested the presence of three MAPKs in T. reesei: Tmk1, Tmk2, and Tmk3. By exploring the phenotypic features of T. reesei tmk3, we first showed elevated NaCl sensitivity and repressed transcription of genes involved in glycerol/trehalose biosynthesis under higher osmolarity, suggesting Tmk3 participates in high osmolarity resistance via derepression of genes involved in osmotic stabilizer biosynthesis. We also showed significant downregulation of genes encoding chitin synthases and a -1,3-glucan synthase, decreased chitin content, budded hyphal appearance typical to cell wall defective strains, and increased sensitivity to calcofluor white/Congo red in the tmk3 deficient strain, suggesting Tmk3 is involved in cell wall integrity maintenance in T. reesei. We further observed the decrease of cellulase transcription and production in T. reesei tmk3 during submerged cultivation, as well as the presence of MAPK phosphorylation sites on known transcription factors involved in cellulase regulation, suggesting Tmk3 is also involved in the regulation of cellulase production. Finally, the expression of cell wall integrity related genes, the expression of cellulase coding genes, cellulase production and biomass accumulation were compared between T. reesei tmk3 grown in solid state media and submerged media, showing a strong restoration effect in solid state media from defects resulted from tmk3 deletion. These results showed novel physiological processes that fungal Hog1-type MAPKs are involved in, and present the first experimental investigation of MAPK signaling pathways in T. reesei. Our observations on the restoration effect during solid state cultivation suggest that T. reesei is evolved to favor solid state growth, bringing up the proposal that the submerged condition normally used during investigations on fungal physiology might be misleading.

Summary:

Introduction: Cells sense their surrounding environments and react to external signals via signal transduction pathways. Among all known signal transducers, the mitogen-activated protein kinase (MAPK) pathway is ubiquitous in almost all eukaryotic species and is one of the most well characterized pathways. This pathway features a signal relay cascade in which three kinates are involved: the MAPK kin as kin (MAPKKK), the MAK kin as kin and MAPK.

Results and Discussion: Phylogenetic analysis of Tmk3 from T. reeseiand other previously characterized MAPKs showed Tm3 apparently cluster with Hog1-type MAPK from other species leading to the suggestion that TmK3likely encodes a Hog type MAPK. Sequence comparison between Tm1 from S. cerevisiaeand Hog3 fromT. euphorbiaeshowed a sequence identity of 66, further supporting this suggestion.

Conclusions: The Hog1-type MAPK Tmk3 in T. reesei is involved in high osmolarity resistance, cell wall integrity maintenance and cellulase production. It is homologous to Hog 1p from S. cerevisiae. Our results suggest that TmK3 functions in the cell wall Integrity signaling pathway similarly to Slt2-type MAKs. We further observed that deletion of Tmkk3leads to an apparent decrease in cellullase transcription, and suggested the involvement of Tk3in cellulose generation regulation by phosphorylating transcription factors.

Figures:

Figure 1: Growth of T. reesei parent strain and T. reesei tmk3 on plates.

Figure 2: Biomass accumulations in submerged media.

Figure 3: Test of sensitivity to NaCl, CR and CFW.

Figure 4: Response of transcriptional levels of glycerol and trehalose biosynthesis genes to elevated osmolarity.

Figure 5: Microscopic images of T. reesei parent strain and T. reesei tmk3 hyphae.

Figure 6: Transcriptional changes of chitin synthase- and -1,3-glucan synthase-coding genes.

Figure 7: Production of extracellular proteins, cellulases and hemicellulase in submerged media.

Figure 8: Transcriptional changes of cellulase- and hemicellulase-coding genes.

Figure 9: Biomass accumulations on solid state media.

Figure 10: Production of extracellular proteins, cellulases and hemicellulase in solid state media.

Mentioned biomedical entities:

Genes: ACEI , BXL1, CR, Cellulase, Chitinase, Cre1, E100, FPA, GPD1, Gna1, Gna3, Hog1, MAPK, MAPKKK, MO, MpkA, Slt2, Tmk1, Tmk2, Tmk3, Xyr1, bgl1, cbh1, cbh2, cellulase, chitinase, egl1, fus3, gpd1, pyr4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. fumigatus, T. reesei

Tissues: tube

Pathways: N/A

Query Result 24/25

Genetic engineering of Trichoderma reesei cellulases and their production

Druzhinina, Irina S.; Kubicek, Christian P.

Microbial Biotechnology

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Keywords Frequency (TF): cellulase: 0.008; reesei: 0.006; cbh1: 0.001; xyr1: 0.000

Abstract:

Lignocellulosic biomass, which mainly consists of cellulose, hemicellulose and lignin, is the most abundant renewable source for production of biofuel and biorefinery products. The industrial use of plant biomass involves mechanical milling or chipping, followed by chemical or physicochemical pretreatment steps to make the material more susceptible to enzymatic hydrolysis. Thereby the cost of enzyme production still presents the major bottleneck, mostly because some of the produced enzymes have low catalytic activity under industrial conditions and/or because the rate of hydrolysis of some enzymes in the secreted enzyme mixture is limiting. Almost all of the lignocellulolytic enzyme cocktails needed for the hydrolysis step are produced by fermentation of the ascomycete Trichoderma reesei (Hypocreales). For this reason, the structure and mechanism of the enzymes involved, the regulation of their expression and the pathways of their formation and secretion have been investigated in T.reesei in considerable details. Several of the findings thereby obtained have been used to improve the formation of the T.reesei cellulases and their properties. In this article, we will review the achievements that have already been made and also show promising fields for further progress.

Summary:

Introduction: Plant biomass is the most abundant renewable source for conversion to biofuel and biorefinery products. It consists of lignocellulose primary made of various polysaccharides and the aromatic polymer ligin, which provide mechanical strength to the plants and render them highly resistant to attack from pathogens (Houstonetal). The main polysoccharides in lignocellulocellulose comprise the glucohomopolysaccharide cellulose (2050, w/w) and hemicelluloses (1535, m/w), which depending on the plantare primarily made up of a xylan, glucuronoxylan,xyloglucan, Glucomannan and arabinoxylan backbones with heterogeneous sidechains (Kubicek,; lvarezetal).

Conclusions: In this article, we have focused on saccharification, and discussed how the production and properties of the cellulases from the major industrial producerT.reeseicould be changed towards a better process economy. In this regard, it is interesting to note that despite the amount of work dedicated to the understanding of the catalytic mechanisms of some of the major celllullases comparatively little work has been published that shows whether the introduction of activity increasing mutations indeed lead to strains that produce better enzyme preparations.

Figures:

Figure 1: Cartoon summarizing the current knowledge about the Trichoderma reesei enzymes that attack and hydrolyse cellulose. Abbreviations: CEL 7A cellobiohydrolase CBH 1; CEL 6A cellobiohydrolase CBH 2; CBM 1 cellulosebinding domain, if present; EG , endoglucanase; AA 9, lytic polysaccharide monooxygenase; BGL glucosidase. CBH 1 and CBH 2 cleave at the (reducing and nonreducing, respectively) ends of the cellulose chain. EG cleaves in amorphous cellulose regions, AA 9 can act on both crystalline and lesscrystalline regions. The oligosaccharides are further hydrolysed to d glucose by glucosidase (BGL , EC 3.2.1.21).

Figure 2: An overview of genetic engineering strategies that offer new perspectives for the further improvement of Trichoderma reesei cellulase formation. Abbreviations: TF transcription factor, RER rough endoplasmic reticulum.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, BGL1, Bey, C1, CBH1, CBH2, CBM1, CEL12A, CEL1B, CEL6A, CEL7A, CEL7B, CRE1, Cas9, Cellulase, GCN5, GH, GHs, Gal4, K14, LaeA, RCE1, RUT, SWO1, Saka, T, Tel, Tm, VIB1, XYN1, XYR1, ace1, cbh1, cbh2, cellulase, cre1, histone, lae1, pyr4, vel1, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Colletotrichum gloeosporioides, T.reesei

Tissues: N/A

Pathways: N/A

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Enhancing saccharification of wheat straw by mixing enzymes from genetically-modified Trichoderma reesei and Aspergillus niger

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Abstract:
To increase the efficiency of enzymatic hydrolysis for plant biomass conversion into renewable biofuel and chemicals.

Summary:
Results: By overexpressing the point mutation A824V transcriptional activator Xyr1 in Trichoderma reesei, carboxymethyl cellulase, cellobiosidase and -d-glucosidases activities of the best mutant were increased from 1.8IU/ml, 0.1IU/ ml and 0.05IU/l to 4.8IU/Ml, 0.44IU/ Ml and 0.33IU/ML, respectively. The sugar yield of wheat straw saccharification by combining enzymes from this mutant and the Aspergillus niger genetically modified strain creA/xlnRc/araRcwas improved up to 7.5mg/mL, a 229 increase compared to the combination of wild type strains.

Conclusions: Mixing enzymes from T. reesei and A. niger combined with the genetic modification of transcription factors is a promising strategy to increase saccharification efficiency. 2016 Elsevier B.V. All rights reserved. - 2015 Elser et al. . Phys.org. AB. Molecular biology of Saccharomyces cerevisiae. epub ahead of print.

Introduction: Depleting fossil energy sources and related environmental issues have stimulated great interest in alternative renewable biofuel sources. Plant biomass is a promising source of raw material for biofuel production. A key bottleneck for the bioconversion of plant biomass into biofuels is the high cost of enzymes and their insufficient efficiency in releasing sugars from lignocellulose. To achieve cost-effective bioconversations, maximizing the enzyme yield and optimizing enzyme mixtures is considered an attractive approach with great potential. Trichoderma reeseiandAspergillus nigerare established industrial producers of enzyme that degrade plant biomass.

Results and discussion: Overexpression of native and constitutivexyr1leads to significant increase of enzyme yields in T. reesei. The three T.reesian strains: wild type, overexpressedxyl1and overexposed constitutivexxyr1, were grown on minimal medium with 1 carbon source (xylan, xylose, glucose, wheat straw and Avicel). No significant growth improvement between the T. Reesiwild type and Xyr1/Xyl2 mutant strains was observed on glucose, Avicel or wheat straw (Fig. In contrast, on xilan and xlylose, the colony size of the Xxy1 mutants was considerably increased compared to the wild type. This correlates with the previously demonstrated role of Xir1 in xilin and ylin utilization (Stricker et al Furthermore, overexpression of the constitutive xxyR1resulted in better growth than that of nativexyL1) suggesting the point mutation (A824V) in the gene encoding Xr1 enhances the utilization ability of xlin and/or yrl2 by T.

Figures:
Figure 1: Growth profiles of the parental strain (WT), QM9414 uridine auxotrophic mutant; OE xyr1 -1 and OE xyr1 -2, overexpressed xyr1 ; OEm xyr1 , overexpressed constitutive xyr1 (A804V). 1000 spores in 2l ACES were spotted on the center of minimal medium plates with 1% carbon source (W/V, xylan from beechwood, xylose, glucose, wheat straw and Avicel, respectively) and incubated at 30C. Pictures were taken three days (3d) after inoculation. The arrows show the edges of the colonies

Figure 2: Exo-protein profiles of the wild type and mutant strains grown on Avicel and wheat straw. The strains were inoculated at 10 6 spores/ml into 250ml flasks containing 50ml minimal medium with a Avicel PH-101(1%, W/V) or b wheat straw (3%, W/V) and cultivated at 30C, 200rpm. Culture supernatants of the fungal strains were collected after four (4d) and seven (7d) days by centrifugation at 4C and 12,000 g for 10min

Figure 3: Cellulase activities measured from the wild type and the mutant strains grown on Avicel and wheat straw. a CMCase, b p NPCase and c p NPGase activities. The culture condition and the sample treatment were the same as for Fig. 1 . Vertical bars indicate standard deviation of biological duplicates

Figure 4: Saccharification analysis of wheat straw. a Total reducing sugar yield; b Individual sugar concentrations. Wheat straw was filtered, autoclaved in demi-water and washed thoroughly with demi-water to remove the soluble sugars. Error bars indicate standard deviation of biological duplicates

Mentioned biomedical entities:
Genes: AraR, B, Cellulase, STW, XlnR, Xyr1, araR, cellulase, creA, pdc, xlnR, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: T. reesei

Tissues: N/A

Pathways: N/A